

Online Resource 1: Proofs and Mathematical Supplement

Supplementary information for *Pruning Deep Neural Networks via the Marchenko–Pastur Distribution*

Leonid Berlyand

Department of Mathematics
Pennsylvania State University
University Park, PA 16802, USA

Theo Bourdais

Department of Computing and Mathematical Sciences
California Institute of Technology
Pasadena, CA 91125, USA

Houman Owhadi

Department of Computing and Mathematical Sciences
California Institute of Technology
Pasadena, CA 91125, USA

Yitzchak Shmalo*

Department of Mathematics
Pennsylvania State University
University Park, PA 16802, USA

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Corresponding author: Yitzchak Shmalo, yitzchak.shmalo@gmail.com

Caption. This file is Online Resource 1. It contains the proof details, Gaussian and abstract random-matrix specializations, finite-rank deformation material, and mathematical corollaries supporting the main manuscript. The main manuscript cites this file as “Online Resource 1.” It is intended to be reviewed and published online exactly as submitted.

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*Corresponding author: Yitzchak Shmalo. Email: yitzchak.shmalo@gmail.com.

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1 Additional Deterministic Certificate Corollaries

Corollary 1.1 (Weighted elastic-net mask certificate). *Under the hypotheses of Theorem 5.4, let $\Gamma_2 = (\gamma_{2,ij})$ and $\Gamma_1 = (\gamma_{1,ij})$ be nonnegative coordinatewise weights, and define*

$$\mathcal{J}_\Gamma(A) := \mathcal{L}_{\text{CE}}(A) + \sum_{i,j} \gamma_{2,ij} A_{ij}^2 + \sum_{i,j} \gamma_{1,ij} |A_{ij}| + \mathcal{C}.$$

Then, for every $\theta \in [0, 1]$,

$$\mathcal{J}_\Gamma(A_\theta) - \mathcal{J}_\Gamma(A_1) \leq (1 - \theta) B_T(R) - (1 - \theta^2) \sum_{i,j} \gamma_{2,ij} R_{ij}^2 - (1 - \theta) \sum_{i,j} \gamma_{1,ij} |R_{ij}|.$$

In particular, full pruning is certified to decrease the weighted elastic-net objective whenever

$$B_T(R) < \sum_{i,j} \gamma_{2,ij} R_{ij}^2 + \sum_{i,j} \gamma_{1,ij} |R_{ij}|.$$

Corollary 1.2 (Weighted stationarity controls removable masks). *Under the hypotheses of Corollary 1.1, assume the one-sided directional stationarity condition*

$$\liminf_{h \downarrow 0} \frac{\mathcal{J}_\Gamma(A_{1-h}) - \mathcal{J}_\Gamma(A_1)}{h} \geq -\varepsilon$$

holds for some $\varepsilon \geq 0$. Then

$$2 \sum_{i,j} \gamma_{2,ij} R_{ij}^2 + \sum_{i,j} \gamma_{1,ij} |R_{ij}| \leq B_T(R) + \varepsilon.$$

More generally, if

$$\mathcal{J}_\Gamma(A_\theta) \geq \mathcal{J}_\Gamma(A_1) - \varepsilon(1 - \theta) \quad \forall \theta \in [0, 1],$$

then for every $\theta < 1$,

$$(1 + \theta) \sum_{i,j} \gamma_{2,ij} R_{ij}^2 + \sum_{i,j} \gamma_{1,ij} |R_{ij}| \leq B_T(R) + \varepsilon.$$

Corollary 1.3 (Deterministic multi-layer margin stability). *Use the ordered pruning sequence and notation of Theorem 5.8. Let $z^{(j)}(s)$ be the logit vector after the j -th mask operation, with $z^{(0)}$ denoting the original network and $z^{(M)}$ the final pruned network. Suppose that for each $s \in T$ and each step j there is a deterministic bound $\eta_{j,s} \geq 0$ such that*

$$\|z^{(j)}(s) - z^{(j-1)}(s)\|_\infty \leq \eta_{j,s}.$$

Define $\eta_s^{\text{tot}} := \sum_{j=1}^M \eta_{j,s}$. Then

$$|\mathcal{L}_{\text{CE}}(\alpha^{(M)}) - \mathcal{L}_{\text{CE}}(\alpha^{(0)})| \leq \frac{2}{|T|} \sum_{s \in T} \eta_s^{\text{tot}}.$$

Moreover, if $p_0(s) \in \arg \max_j z_j^{(0)}(s)$ and

$$\Delta_0^{\text{pred}}(s) := z_{p_0(s)}^{(0)}(s) - \max_{j \neq p_0(s)} z_j^{(0)}(s),$$

then the predicted label at s can change between the original and final networks only if

$$\Delta_0^{\text{pred}}(s) \leq 2\eta_s^{\text{tot}}.$$

For the true-label margin

$$\Delta_0^{\text{true}}(s) := z_{C(s)}^{(0)}(s) - \max_{j \neq C(s)} z_j^{(0)}(s),$$

the accuracy change obeys

$$|\text{acc}_{\alpha^{(M)}}(T) - \text{acc}_{\alpha^{(0)}}(T)| \leq \frac{1}{|T|} \#\{s \in T : |\Delta_0^{\text{true}}(s)| \leq 2\eta_s^{\text{tot}}\}.$$

Corollary 1.4 (Sub-Gaussian sufficient condition for the data-path budget). *Let $v_s = \psi_1(s)$, and assume L_s are nonnegative upper bounds for the post-layer Lipschitz constants, deterministic or measurable with respect to variables independent of R . Let the rows $r_1, \dots, r_n$ of R be independent and centered conditionally on those variables, and suppose that for some $c, g > 0$,*

$$\mathbb{P}(|\langle r_i, v \rangle| > t \mid v) \leq 2 \exp\left(-\frac{cnt^2}{g\|v\|_2^2}\right)$$

for all i, v , and $t > 0$. Then, with probability at least $1 - \delta$,

$$B_T(R) \leq \frac{2}{|T|} \sum_{s \in T} L_s \|v_s\|_2 \sqrt{\frac{g}{cn} \log \frac{2n|T|}{\delta}}.$$

Corollary 1.5 (Simultaneous multi-layer Gaussian data-path budget). *Consider the ordered finite pruning sequence in Theorem 5.8. At step j , let the removed component be modeled by $R_j \in \mathbb{R}^{n_j \times m_j}$ with independent $N(0, g_j/n_j)$ entries, conditionally independent of the intermediate pre-layer vectors $v_{j,s}$ and Lipschitz bounds $L_{j,s}$. Let $\delta_1, \dots, \delta_M > 0$ satisfy $\sum_{j=1}^M \delta_j \leq \delta$. Then, with probability at least $1 - \delta$, all layerwise budgets satisfy*

$$B_j(R_j) \leq \frac{2}{|T|} \sum_{s \in T} L_{j,s} \|v_{j,s}\|_2 \sqrt{\frac{2g_j}{n_j} \log \frac{2n_j|T|}{\delta_j}} \quad (1 \leq j \leq M).$$

Consequently, on the same event, Theorem 5.8 holds with $\sum_j B_j(R_j)$ replaced by the sum of the displayed right-hand sides.

Corollary 1.6 (Simultaneous multi-layer sub-Gaussian data-path budget). *Use the ordered pruning sequence and notation of Corollary 1.5, but at each step j replace the Gaussian model by the row-concentration hypothesis of Corollary 1.4 with (R, n, c, g, v_s, L_s) replaced by $(R_j, n_j, c_j, g_j, v_{j,s}, L_{j,s})$. Let $\delta_1, \dots, \delta_M > 0$ satisfy $\sum_{j=1}^M \delta_j \leq \delta$. Then, with probability at least $1 - \delta$,*

$$B_j(R_j) \leq \frac{2}{|T|} \sum_{s \in T} L_{j,s} \|v_{j,s}\|_2 \sqrt{\frac{g_j}{c_j n_j} \log \frac{2n_j|T|}{\delta_j}} \quad (1 \leq j \leq M).$$

Consequently, the same deterministic multi-layer certificate applies with the displayed sub-Gaussian bounds.

2 Notes on the Abstract Additive Framework

2.1 Relocated abstract additive corollaries

Corollary 2.1 (Growing-set abstract loss reduction). *Let T_N be a finite labeled set that may depend on N . Let L_{reg, T_N} denote cross-entropy averaged over T_N , and let L_{T_N} denote the corresponding regularized objective with the same Frobenius penalty as in (7). Assume Assumptions 4.1–4.2,*

$$\langle S, R \rangle_F = o_{\mathbb{P}}(1), \quad |T_N|d_2(N) \rightarrow 0,$$

and

$$\frac{1}{|T_N|} \sum_{s \in T_N} a_{\phi}(N, s) \rightarrow 0$$

in probability. Then

$$L_{T_N}(\alpha_S) = L_{T_N}(\alpha_W) - \lambda_{\text{reg}} \|R\|_F^2 + o_{\mathbb{P}}(1).$$

Corollary 2.2. *Under the hypotheses of Theorem 4.5, for any fixed $\vartheta \in [0, 1]$, define $W(\vartheta) = S + \vartheta R$. Then*

$$L(\alpha_{W(\vartheta)}) = L(\alpha_W) - \lambda_{\text{reg}}(1 - \vartheta^2) \|R\|_F^2 + o_{\mathbb{P}}(1) \quad (N \rightarrow \infty).$$

The same asymptotic statement holds along any deterministic sequence $\vartheta_N \in [0, 1]$.

Corollary 2.3 (Finite- N scaled loss bound along the denoising path). *Assume the hypotheses of Theorem 4.5, and define $W(\vartheta) = S + \vartheta R$ for $\vartheta \in [0, 1]$. Then for every $\eta > 0$ there exists an event $E_{N, \eta}$ with*

$$\mathbb{P}(E_{N, \eta}) \geq 1 - |T|d_2(N) - \mathbb{P}(|\langle S, R \rangle_F| > \eta)$$

such that on $E_{N, \eta}$,

$$L(\alpha_{W(\vartheta)}) \leq L(\alpha_W) - \lambda_{\text{reg}}(1 - \vartheta^2) \|R\|_F^2 + (1 - \vartheta) \frac{2}{|T|} \sum_{s \in T} a_{\phi}(N, s) + 2\lambda_{\text{reg}}(1 - \vartheta)\eta$$

for every $\vartheta \in [0, 1]$.

Corollary 2.4 (Eventual supercriticality of persistent spike clusters in a BGN-type deformation regime). *Under the hypotheses of Theorem 4.7, assume additionally that for each N , $W^*(N) = S^*(N) + R^*(N)$ lies in a finite-rank deformation regime satisfying all hypotheses of Theorem 5.4 for the spikes considered below. Assume the regime has deterministic bulk edge $b_+(N)$, critical threshold $\theta_c(N)$, strictly decreasing outlier transform*

$$D_N : (b_+(N), \infty) \rightarrow (0, D_N(b_+(N)^+)),$$

local Lipschitz/invertibility intervals around the outliers, and concentration of the relevant sample singular values in those intervals. Assume $\theta_c(N) \rightarrow 0$, $b_+(N) \rightarrow 0$, and that for some fixed $k \geq 1$ there exist constants $\bar{\sigma}_1 > \dots > \bar{\sigma}_k > 0$ such that $\sigma_i^*(N) \rightarrow \bar{\sigma}_i$ in probability for each $1 \leq i \leq k$. Then:

1. $\mathbb{P}(\sigma_i^*(N) > \theta_c(N)) \rightarrow 1$ for each $1 \leq i \leq k$;
2. these k spikes are eventually supercritical, so the corresponding singular values of $W^*(N)$ form outlier clusters above $b_+(N)$. If the i -th spike is separated from the other nonzero singular values of $S^*(N)$ by a gap $\Delta_i(N) > 0$ with $\|R^*(N)\|_{\text{op}}/\Delta_i(N) \rightarrow 0$ in probability, then the associated rank-one left and right spectral projectors converge in operator norm to the corresponding projectors of $S^*(N)$;

3. defining $\hat{\sigma}_i^{(D)}(N) := D_N(s_i(W^*(N)))^{-1/2}$ on the event $s_i(W^*(N)) > b_+(N)$, whose probability tends to one, and arbitrarily off that event, one has $\hat{\sigma}_i^{(D)}(N) - \sigma_i^*(N) \rightarrow 0$ in probability.

Assumptions 4.1–4.2 are the abstract data-path version of the Gaussian conditions in Appendix 3; Assumption 3.3 gives the iid-Gaussian sufficient condition, while the optional ESD condition is used only for spectral conclusions.

Example 2.5. The square iid-Gaussian case with variance $1/N$ gives Assumption 4.2 with $J(v) = \|v\|_2$, $d_1(N) = 2\sqrt{2\log(2N)/N}$, and $d_2(N) = 1/N$; see Corollary 5.10. The conditional formulation also accommodates sub-Gaussian rows, orthogonally invariant perturbations, or moderate training-induced dependence when the concentration estimate survives.

Example 2.6 (Sub-Gaussian rows). The row-concentration variant in Corollary 1.4 gives the corresponding Assumption 4.2 sufficient condition.

Assumption 4.3 is used only for spectral conclusions. Loss reduction uses the data-path smallness $a_\phi(N, s) \rightarrow 0$ and the cross-term condition; stationarity-collapse results additionally use the stated one-sided stationarity hypothesis.

Remark 2.7 (Regularisation scaling). Throughout the loss-reduction statements, the elastic-net coefficients λ_1, λ_2 are treated as specified constants. An iid-Gaussian admissible component $R \in \mathbb{R}^{N \times N}$ with variance g/N has $\|R\|_F^2 = \Theta(gN)$ in expectation, so the loss-reduction term $\lambda_2\|R\|_F^2$ scales as $\Theta(\lambda_2 gN)$. The certificate inequality of Theorem 5.4 holds for any specified coefficient scaling, including coefficients normalized by parameter count or layer width. The data-path budget $B_T(R)$ is independent of λ_1, λ_2 , while the elastic-net reward is not; only the numerical certificate margin changes with regularizer scale.

Remark 2.8 (Beyond pure Frobenius regularization). The quadratic penalty in (7) supplies a coercive lower bound of order $\|R\|_F^2$. For a differentiable μ -strongly convex regularizer Ω ,

$$\Omega(S + R) - \Omega(S) = \langle \nabla \Omega(S), R \rangle + D_\Omega(S + R, S), \quad D_\Omega(S + R, S) \geq \frac{\mu}{2} \|R\|_F^2.$$

The extension is valid when the first-order term is controlled or negligible. In the Frobenius case this term is $2\langle S, R \rangle_F$. For elastic net, choose $\xi \in \partial\|S\|_1$ and assume $\langle \mu_1 \xi + 2\mu_2 S, R \rangle = o_{\mathbb{P}}(1)$; the quadratic part supplies the strong convexity.

The stationarity theorem assumes only one-sided approximate stationarity at $\vartheta = 1$; exact local minimality is sufficient but stronger. Without the no-atom condition $\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0$, the same proof gives limsup/liminf threshold bounds rather than exact convergence. Corollary 2.4 is the corresponding abstract BBP persistence statement.

Dependency map: deterministic mask certificates use the support condition

$$S = M \odot A, \quad R = (1 - M) \odot A,$$

so $S \odot R = 0$. Abstract additive loss statements replace that support condition by the relevant cross-term control, such as $\langle S, R \rangle_F = o_{\mathbb{P}}(1)$. Random budget corollaries provide sufficient conditions for $B_T(R)$.

Remark 2.9 (Local Lipschitz constants L_s). The constants L_s in (8) are local post-layer ℓ_∞ -Lipschitz bounds; for ViTs this paper treats them as architecture-dependent and uses the empirical proxy audit in Online Resource 2.

3 Gaussian Specialization and Spectral Interpretation

The deterministic mask certificates in Section 5 are closest to the implemented unstructured pruning operation; this appendix gives Gaussian sufficient conditions and spectral interpretations for admissible random-like components.

3.1 Assumptions for the Gaussian specialization

The finite- T convention from Section 4 remains in force unless stated otherwise.

Whenever λ_+ denotes the right MP edge for the eigenvalues of

$$X_\ell = \frac{1}{N} W_\ell^\top W_\ell,$$

we write

$$\sigma_+ := \sqrt{N\lambda_+}$$

for the corresponding edge on the singular-value scale.

Assumption 3.1 (Architecture and outer-layer scaling). We consider the simplified three-layer network

$$X(\alpha, s) = \lambda \circ W_3 \circ \lambda \circ W_2 \circ \lambda \circ W_1 s, \quad s \in T, \quad (1)$$

$$\phi(\alpha, s) = \rho \circ X(\alpha, s), \quad (2)$$

where λ is the coordinatewise absolute value or ReLU, and W_1, W_2, W_3 are the weight matrices. Biases are omitted for simplicity.

For fixed training time t , define

$$a(N, s) := \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}}. \quad (3)$$

We assume that for every $s \in T$, $a(N, s) \rightarrow 0$ as $N \rightarrow \infty$. The same assumption is imposed at any admissible limit point considered below.

Remark 3.2 (Interpretation of Assumption 3.1). This is the concrete version of the abstract factorization Assumption 4.1. The target layer is W_2 ; the computations before W_2 play the role of ψ_1 , and the computations after W_2 play the role of ψ_2 . The quantity $a(N, s)$ is the Gaussian-model version of $a_\phi(N, s) = h_\phi(s) d_1(N)$: it is the logit perturbation scale predicted by multiplying an $N^{-1/2} \sqrt{\log N}$ -size random effect through the outer layer. The assumption $a(N, s) \rightarrow 0$ is therefore the concrete data-path smallness condition needed for the loss and margin bounds.

Assumption 3.3 (Admissible Gaussian decomposition of the middle layer). For each training time t , the middle layer $W_2(t) \in \mathbb{R}^{N \times N}$ admits a decomposition

$$W_2(t) = R_2(t) + S_2(t),$$

where:

1. $R_2(t)$ has i.i.d. centered Gaussian entries

$$R_2(t)_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g(t)}{N}\right), \quad 0 \leq g(t) \leq 1;$$

where $g(t)$ is a deterministic variance scale;

2. $R_2(t)$ is independent of $S_2(t)$;
3. $R_2(t)$ is also independent of the outer-layer parameters $W_1(t)$ and $W_3(t)$.

Remark 3.4. The variance scaling $g(t)/N$ ensures that $R_2(t)$ has bounded operator norm as $N \rightarrow \infty$. Rectangular variants $W_2(t) \in \mathbb{R}^{N \times M}$ with $M/N \rightarrow c \in (0, \infty)$ can be treated similarly.

Remark 3.5 (Interpretation of Assumption 3.3). This is a stylized additive-noise model, not a claim that the implemented ViT mask literally samples an independent Gaussian matrix. It is used to prove a sufficient condition: if the removed component behaves like an independent variance- g/N random matrix along the data path, then it satisfies the perturbation bounds required by the abstract theory. In the notation of Assumption 4.2, it gives $J(v) = \|v\|_2$, $d_1(N) \asymp \sqrt{\log N/N}$, and $d_2(N) \asymp 1/N$. The same variance scale has MP singular-value edge $2\sqrt{g}$ in the square case, which is why the fitted MP edge is a natural diagnostic in the algorithms.

Assumption 3.6 (Sub-root- N structured component for the loss theorems). For each training time t , the structured component $S_2(t)$ satisfies

$$\|S_2(t)\|_F \leq B_N, \quad B_N = o(\sqrt{N}),$$

for some positive deterministic sequence B_N independent of t . Equivalently, the normalized size condition $\|S_2(t)\|_F^2/N \rightarrow 0$ is enough. The loss-reduction theorems use this assumption only to prove

$$\langle S_2(t), R_2(t) \rangle_F = o_{\mathbb{P}}(1);$$

one may replace it directly by this cross-term condition whenever it is known by another argument.

Remark 3.7 (Interpretation of Assumption 3.6). This assumption is a convenient Gaussian sufficient condition for the cross-term bound $\langle S_2, R_2 \rangle_F = o_{\mathbb{P}}(1)$. Conditional on S_2 ,

$$\text{Var}(\langle S_2, R_2 \rangle_F \mid S_2) = \frac{g(t)}{N} \|S_2\|_F^2.$$

Thus $B_N = o(\sqrt{N})$ forces this variance to vanish; this is the Gaussian counterpart of the explicit cross-term hypothesis used in Section 4.

Assumption 3.8 (Approximate low rank for the spectral corollaries). Whenever a spectral conclusion is invoked, we additionally assume that the limit structured component $S_2^*(N)$ is approximately low rank in the sense that there exists a sequence $k_N = o(N)$ such that

$$\sum_{i > k_N} s_i(S_2^*(N))^2 \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

The exact fixed-rank regime is the special case $k_N \equiv r$.

Remark 3.9 (Interpretation of Assumption 3.8). This is the Gaussian specialization of Assumption 4.3 and is used only by the spectral corollaries.

Remark 3.10 (No spectral-separation assumption). Assumptions 3.6–3.8 do *not* require the informative singular values of S_2 to lie above the MP edge σ_+ . Thus signal directions may still be embedded in the bulk created by R_2 .

Definition 3.11 (Admissible decomposition). A decomposition $W_2 = S_2 + R_2$ is called *admissible* if it satisfies Assumptions 3.3–3.6. At a limit point $W_2^*(N)$, an admissible decomposition $W_2^*(N) = S_2^*(N) + R_2^*(N)$ additionally requires that the outer-layer scaling condition from Assumption 3.1 holds for the corresponding limit outer layers $W_1^*(N), W_3^*(N)$, i.e.

$$a_*(N, s) := \|W_3^*(N)\|_{\infty} \|W_1^*(N)s\|_2 \sqrt{\frac{\log(2N)}{N}} \rightarrow 0 \quad \text{for each } s \in T.$$

A *spectrally admissible* decomposition is an admissible decomposition that also satisfies Assumption 3.8.

Definition 3.12 (Local-path convergence of an admissible decomposition). An admissible path decomposition

$$W_2(t, N) = S_2(t, N) + R_2(t, N)$$

is said to converge locally to an admissible limit decomposition

$$W_2^*(N) = S_2^*(N) + R_2^*(N)$$

if, for each fixed N ,

$$\|S_2(t, N) - S_2^*(N)\|_F + \|R_2(t, N) - R_2^*(N)\|_F \rightarrow 0 \quad \text{in probability as } t \rightarrow \infty.$$

3.2 Key perturbation lemma

Lemma 3.13. *Suppose Assumptions 3.1 and 3.3 hold. Define*

$$\tau_N(s) := 2\sqrt{2} \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}} = 2\sqrt{2} a(N, s).$$

If we replace $W_2 = S_2 + R_2$ by S_2 , then

$$\mathbb{P}(\|X(s, \alpha_{S_2}) - X(s, \alpha_{W_2})\|_\infty \leq \tau_N(s)) \geq 1 - \frac{1}{N}.$$

Proof. See Appendix 4, Proof of Lemma 3.13.

For later use, define the classification margin on a finite test set T' :

$$\delta X(s, \alpha) := X_{C(s)}(s, \alpha) - \max_{j \neq C(s)} X_j(s, \alpha),$$

and the corresponding accuracy

$$\text{acc}_\alpha(T') := \frac{1}{|T'|} \#\{s \in T' : \delta X(s, \alpha) > 0\}.$$

Corollary 3.14 (Margin-sensitive label stability). *Suppose Assumptions 3.1 and 3.3 hold. Let T' be a finite test set. Then*

$$\mathbb{P}\left(\left|\text{acc}_{\alpha_{W_2}}(T') - \text{acc}_{\alpha_{S_2}}(T')\right| \leq \frac{1}{|T'|} \#\left\{s \in T' : |\delta X(s, \alpha_{W_2})| \leq 2\tau_N(s)\right\}\right) \geq 1 - \frac{|T'|}{N}.$$

In particular, if

$$\min_{s \in T'} |\delta X(s, \alpha_{W_2})| > 2 \max_{s \in T'} \tau_N(s),$$

then

$$\text{acc}_{\alpha_{W_2}}(T') = \text{acc}_{\alpha_{S_2}}(T')$$

with probability at least $1 - \frac{|T'|}{N}$.

Proof. See Appendix 5, Proof of Corollary 3.14.

3.3 A theorem on how removing randomness reduces the regularized loss

We work with the ℓ_2 -regularized loss

$$L(\alpha(t)) = -\frac{1}{|T|} \sum_{s \in T} \log(\phi_{C(s)}(s, \alpha(t))) + \lambda_{\text{reg}} \sum_{i=1}^L \|W_i(t)\|_F^2, \quad \lambda_{\text{reg}} > 0. \quad (4)$$

We first isolate the unregularized perturbation term.

Lemma 3.15. *Suppose Assumptions 3.1–3.3 hold. Define the unregularized cross-entropy*

$$L_{\text{reg}}(\alpha) := -\frac{1}{|T|} \sum_{s \in T} \log(\phi_{C(s)}(s, \alpha)).$$

Then there exists N_0 such that for all $N \geq N_0$,

$$\mathbb{P}\left(\left|L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})\right| \leq \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a(N, s)\right) \geq 1 - \frac{|T|}{N}.$$

In particular,

$$L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2}) \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

Proof. See Appendix 4, Proof of Lemma 3.15.

Theorem 3.16 (Confidence-parameter finite-sample perturbation bound). *Suppose Assumptions 3.1 and 3.3 hold, and fix a finite set T_0 and $\delta \in (0, 1)$. Define*

$$\tau_{N,\delta}(s) := \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{2g(t) \log(2N|T_0|/\delta)}{N}}.$$

Then, with probability at least $1 - \delta$,

$$\|X(s, \alpha_{S_2}) - X(s, \alpha_{W_2})\|_\infty \leq \tau_{N,\delta}(s) \quad \forall s \in T_0.$$

Consequently, for $T_0 = T$,

$$\left|L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})\right| \leq \frac{2}{|T|} \sum_{s \in T} \tau_{N,\delta}(s)$$

with probability at least $1 - \delta$. If Assumption 3.6 also holds, then for every $\eta > 0$,

$$\mathbb{P}\left(L(\alpha_{S_2}) \leq L(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 + \frac{2}{|T|} \sum_{s \in T} \tau_{N,\delta}(s) + 2\lambda_{\text{reg}} \eta\right) \geq 1 - \delta - 2e^{-N\eta^2/(2B_N^2)}.$$

Proof. See Appendix 4, Proof of Theorem 3.16.

Corollary 3.17 (Growing-set Gaussian loss bound). *Suppose Assumptions 3.1–3.3 hold with a finite labeled set T_N that may depend on N , and let $\delta_N \in (0, 1)$ with $\delta_N \rightarrow 0$. Let L_{reg, T_N} denote cross-entropy averaged over T_N , and let L_{T_N} denote the corresponding regularized objective with the same Frobenius penalty as in (4). Define*

$$b_{N,\delta_N}(T_N) := \frac{2}{|T_N|} \sum_{s \in T_N} \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{2g(t) \log(2N|T_N|/\delta_N)}{N}}.$$

If $b_{N,\delta_N}(T_N) \rightarrow 0$ in probability, then the cross-entropy loss averaged over T_N satisfies

$$L_{\text{reg},T_N}(\alpha_{S_2}) - L_{\text{reg},T_N}(\alpha_{W_2}) \rightarrow 0 \quad \text{in probability.}$$

If Assumption 3.6 also holds, then

$$L_{T_N}(\alpha_{S_2}) \leq L_{T_N}(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 + o_{\mathbb{P}}(1).$$

Proof. The bound follows by applying Theorem 3.16 to the finite set T_N with failure probability δ_N ; the stated assumptions $\delta_N \rightarrow 0$ and $b_{N,\delta_N}(T_N) \rightarrow 0$ give the result.

Theorem 3.18. Suppose Assumptions 3.1–3.6 hold. If $W_2 = R_2 + S_2$ and we replace W_2 by S_2 , then for every fixed training time t and every $\eta > 0$,

$$\mathbb{P}\left(L(\alpha_{S_2}) \leq L(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 + \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a(N, s) + 2\lambda_{\text{reg}} \eta\right) \geq 1 - \frac{|T|}{N} - 2e^{-N\eta^2/(2B_N^2)}.$$

In particular,

$$L(\alpha_{S_2}) = L(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 + o_{\mathbb{P}}(1) \quad (N \rightarrow \infty).$$

Proof. See Appendix 5, Proof of Theorem 3.18.

Corollary 3.19. Suppose Assumptions 3.1–3.6 hold. For any $\vartheta \in [0, 1]$, define

$$W_2(\vartheta) = S_2 + \vartheta R_2.$$

Then for every fixed training time t and every $\eta > 0$, there is an event $E_{N,\eta}$ with

$$\mathbb{P}(E_{N,\eta}) \geq 1 - \frac{|T|}{N} - 2e^{-N\eta^2/(2B_N^2)}$$

such that on $E_{N,\eta}$, for every $\vartheta \in [0, 1]$,

$$L(\alpha_{W_2(\vartheta)}) \leq L(\alpha_{W_2}) - \lambda_{\text{reg}}(1 - \vartheta^2) \|R_2\|_F^2 + \frac{4\sqrt{2}(1 - \vartheta)}{|T|} \sum_{s \in T} a(N, s) + 2\lambda_{\text{reg}}(1 - \vartheta)\eta.$$

In particular, for each fixed $\vartheta \in [0, 1]$,

$$L(\alpha_{W_2(\vartheta)}) = L(\alpha_{W_2}) - \lambda_{\text{reg}}(1 - \vartheta^2) \|R_2\|_F^2 + o_{\mathbb{P}}(1) \quad (N \rightarrow \infty).$$

The same asymptotic statement holds along any deterministic sequence $\vartheta_N \in [0, 1]$.

Proof. See Appendix 5, Proof of Corollary 3.19.

Theorem 3.20 (Entrywise thresholding in a sparse signal-plus-noise model). Let $W_N = S_N + R_N \in \mathbb{R}^{N \times M_N}$, where R_{ij} are independent $N(0, g_N/N)$ random variables, $g_N \geq 0$, and S_N is deterministic or independent of R_N . Fix $\delta \in (0, 1)$ and set

$$t_{N,\delta} := \sqrt{\frac{2g_N \log(2NM_N/\delta)}{N}}.$$

Define the entrywise hard-thresholded matrix

$$\mathcal{H}_t(W)_{ij} := W_{ij} \mathbf{1}_{\{|W_{ij}| > t\}}.$$

Then, with probability at least $1 - \delta$,

$$\{(i, j) : |(S_N)_{ij}| > 2t_{N,\delta}\} \subseteq \text{supp}(\mathcal{H}_{t_{N,\delta}}(W_N)) \subseteq \text{supp}(S_N).$$

Equivalently, thresholding creates no false positives and can miss only entries of S_N whose magnitude is at most $2t_{N,\delta}$:

$$\text{supp}(S_N) \setminus \text{supp}(\mathcal{H}_{t_{N,\delta}}(W_N)) \subseteq \{(i, j) : 0 < |(S_N)_{ij}| \leq 2t_{N,\delta}\}.$$

Let $\mathcal{B}_{N,\delta}$ be the beta-min event

$$\mathcal{B}_{N,\delta} := \left\{ \min_{(i,j) \in \text{supp}(S_N)} |(S_N)_{ij}| > 2t_{N,\delta} \right\},$$

with the condition interpreted as vacuous if $\text{supp}(S_N) = \emptyset$. If $\mathbb{P}(\mathcal{B}_{N,\delta}) \geq 1 - \gamma$, then entrywise thresholding W_N at level $t_{N,\delta}$ recovers the support of S_N exactly with probability at least $1 - \delta - \gamma$:

$$\text{supp}(\mathcal{H}_{t_{N,\delta}}(W_N)) = \text{supp}(S_N).$$

In particular, if $\mathcal{B}_{N,\delta}$ holds deterministically, or almost surely when S_N is random, the recovery probability is at least $1 - \delta$. This result is a stylized support-recovery statement. Without beta-min separation it still gives a no-false-positive and large-signal-retention certificate, but exact support recovery requires all nonzero structured entries to clear the $2t_{N,\delta}$ noise buffer. It is not a proof that the ViT pruning masks identify the true structured support.

Proof. See Appendix 4, Proof of Theorem 3.20.

3.4 Asymptotic magnitude of admissible noise

We now replace exact local minimality by a weaker one-sided stationarity condition along the denoising direction $R_2^*(N)$.

Theorem 3.21. *Suppose Assumptions 3.1–3.6 hold along training. Assume that:*

1. *for each N , the parameter vector $\alpha(t, N)$ converges in probability as $t \rightarrow \infty$ to a limit $\alpha^*(N)$;*
2. *the limiting middle layer $W_2^*(N)$ admits an admissible decomposition*

$$W_2^*(N) = S_2^*(N) + R_2^*(N)$$

in the sense of Definition 3.11.

For $\vartheta \in [0, 1]$, let $\alpha_\vartheta^*(N)$ denote the parameter vector obtained from $\alpha^*(N)$ by replacing $W_2^*(N)$ with $S_2^*(N) + \vartheta R_2^*(N)$. Assume further that there exists a sequence of nonnegative random variables $\epsilon_N = o_{\mathbb{P}}(1)$ such that the one-sided directional stationarity event

$$\liminf_{h \downarrow 0} \frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \geq -\epsilon_N$$

has probability tending to one. Define

$$\bar{a}_N := \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a_*(N, s).$$

Then

$$\|R_2^*(N)\|_F^2 \leq \frac{\bar{a}_N + \epsilon_N}{2\lambda_{\text{reg}}} + o_{\mathbb{P}}(1).$$

In particular, if $\bar{a}_N + \epsilon_N \rightarrow 0$ in probability, then for every $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \mathbb{P}(\|R_2^*(N)\|_F \leq \epsilon) = 1.$$

If, in addition, there exists an admissible path decomposition

$$W_2(t, N) = S_2(t, N) + R_2(t, N)$$

which converges locally to the limit decomposition in the sense of Definition 3.12, then for every $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \lim_{t \rightarrow \infty} \mathbb{P}(\|R_2(t, N)\|_F \leq \epsilon) = 1.$$

Proof. See Appendix 5, Proof of Theorem 3.21.

Remark 3.22. Theorem 3.21 requires only one-sided directional stationarity at $\vartheta = 1$; a genuine local minimum gives $\epsilon_N \equiv 0$. Its second statement transfers the limit-point conclusion to any admissible training-path decomposition converging locally to the limit. Since the Gaussian $\|R_2^*(N)\|_F$ has no atom at any $\epsilon > 0$, pathwise convergence in probability yields convergence of the corresponding distribution functions at those thresholds.

Corollary 3.23 (Variance-scale collapse at the limit point). *Under the hypotheses of Theorem 3.21, write the admissible Gaussian limit perturbation as*

$$R_2^*(N)_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g_*(N)}{N}\right).$$

Here $g_*(N)$ is the deterministic variance scale of the limit perturbation. Then

$$g_*(N) N \rightarrow 0.$$

In particular, in the square case the MP singular-value edge of the admissible limit perturbation satisfies

$$\sigma_{+,*}^{\text{bulk}}(N) = 2\sqrt{g_*(N)} \rightarrow 0.$$

For a rectangular $N \times M_N$ perturbation with entries $N(0, g_*(N)/N)$, the corresponding Frobenius collapse condition is $g_*(N)M_N \rightarrow 0$; when $M_N/N \rightarrow c \in (0, \infty)$, this is equivalent to $g_*(N)N \rightarrow 0$.

Proof. See Appendix 5, Proof of Corollary 3.23.

Corollary 3.24 (Bulk collapse and spike stabilization at the limit point). *Under the hypotheses of Theorem 3.21, assume additionally that the limit decomposition is spectrally admissible in the sense of Definition 3.11. Let*

$$s_1(W_2^*(N)) \geq \cdots \geq s_N(W_2^*(N))$$

denote the singular values of $W_2^*(N)$, and let

$$\sigma_1^*(N) \geq \sigma_2^*(N) \geq \cdots \geq 0$$

denote the singular values of $S_2^*(N)$. Let $k_N = o(N)$ be as in Assumption 3.8. Then:

1. $\|R_2^*(N)\|_{\text{op}} \rightarrow 0$ in probability;

2. for each fixed $i \geq 1$,

$$|s_i(W_2^*(N)) - \sigma_i^*(N)| \rightarrow 0 \quad \text{in probability;}$$

3.

$$\sum_{i > k_N} s_i(W_2^*(N))^2 \rightarrow 0$$

in probability;

4. for every $\epsilon > 0$,

$$\nu_{W_2^*(N)}((\epsilon, \infty)) \rightarrow 0 \quad \text{in probability,}$$

where $\nu_{W_2^*(N)}$ is the singular-value empirical spectral distribution.

If, in addition, $\sigma_i^*(N) \rightarrow \bar{\sigma}_i$ for some constants $\bar{\sigma}_i \geq 0$, then

$$s_i(W_2^*(N)) \rightarrow \bar{\sigma}_i \quad \text{in probability for each fixed } i \geq 1.$$

Proof. See Appendix 5, Proof of Corollary 3.24.

Corollary 3.25 (Singular-subspace stabilization under an asymptotic gap condition). *Under the hypotheses of Theorem 3.21, let*

$$\sigma_1^*(N) \geq \sigma_2^*(N) \geq \cdots \geq 0$$

denote the singular values of $S_2^(N)$. Fix $k \geq 1$, and define*

$$\delta_k(N) := \sigma_k^*(N) - \sigma_{k+1}^*(N),$$

with the convention $\sigma_j^(N) = 0$ for j beyond the rank of $S_2^*(N)$. Assume that $\mathbb{P}(\delta_k(N) > 0) \rightarrow 1$ and that, after setting the ratio below to $+\infty$ on the event $\{\delta_k(N) = 0\}$,*

$$\frac{\|R_2^*(N)\|_{\text{op}}}{\delta_k(N)} \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

Let $U_{,k}(N)$ and $V_{*,k}(N)$ be orthonormal bases for the top- k left and right singular subspaces of $S_2^*(N)$, respectively. Let $\hat{U}_k(N), \hat{V}_k(N) \in \mathbb{R}^{N \times k}$ be matrices whose columns are orthonormal left and right singular vectors corresponding to the top k singular values of $W_2^*(N)$, and define the orthogonal projectors*

$$\begin{aligned} P_{U_{*,k}}(N) &:= U_{*,k}(N)U_{*,k}(N)^\top, & P_{V_{*,k}}(N) &:= V_{*,k}(N)V_{*,k}(N)^\top, \\ \hat{P}_{U,k}(N) &:= \hat{U}_k(N)\hat{U}_k(N)^\top, & \hat{P}_{V,k}(N) &:= \hat{V}_k(N)\hat{V}_k(N)^\top. \end{aligned}$$

Then

$$\|\hat{P}_{U,k}(N) - P_{U_{*,k}}(N)\|_{\text{op}} \rightarrow 0, \quad \|\hat{P}_{V,k}(N) - P_{V_{*,k}}(N)\|_{\text{op}} \rightarrow 0$$

in probability as $N \rightarrow \infty$.

Proof. See Appendix 5, Proof of Corollary 3.25.

Remark 3.26. Corollary 3.25 converts spectral shrinkage into geometric stability: if residual operator-norm noise is asymptotically smaller than $\delta_k(N)$, the corresponding left/right singular subspaces become identifiable. This does not require a fixed spike floor; $\delta_k(N)$ may tend to 0 if it still dominates the noise.

Corollary 3.27 (Eventual Gaussian supercriticality of persistent spikes). *Under the hypotheses of Theorem 3.21, write*

$$R_2^*(N)_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{g_*(N)}{N}\right).$$

Assume additionally that, for this corollary only, the admissible limit decomposition lies in the square Gaussian finite-rank deformation regime of Appendix 5; in particular, $S_2^(N)$ has rank at most r_0 for some deterministic r_0 independent of N . Assume that for some fixed $k \geq 1$ there exist constants*

$$\bar{\sigma}_1 > \cdots > \bar{\sigma}_k > 0$$

such that

$$\sigma_i^*(N) \rightarrow \bar{\sigma}_i \quad \text{in probability as } N \rightarrow \infty$$

for each $1 \leq i \leq k$. Then:

1.

$$\mathbb{P}(\sigma_i^*(N) > \sqrt{g_*(N)}) \rightarrow 1 \quad \text{for each } 1 \leq i \leq k;$$

2. *in the sense of Theorem 5.3, these k spikes are eventually supercritical, so the corresponding singular values of $W_2^*(N)$ form BBP outlier clusters above the Gaussian bulk edge $2\sqrt{g_*(N)}$. Moreover, if the i -th spike is separated from the other nonzero singular values of $S_2^*(N)$ by a gap $\Delta_i(N) > 0$ with*

$$\frac{\|R_2^*(N)\|_{\text{op}}}{\Delta_i(N)} \rightarrow 0 \quad \text{in probability,}$$

then the rank-one left and right spectral projectors of $W_2^(N)$ associated with this spike converge in operator norm to the corresponding projectors of $S_2^*(N)$;*

3. *defining the inverse-BBP estimator on the event $s_i(W_2^*(N))^2 > 4g_*(N)$, whose probability tends to one, by*

$$\hat{\sigma}_i^{\text{BBP}}(N) := \frac{1}{2} \left(s_i(W_2^*(N)) + \sqrt{s_i(W_2^*(N))^2 - 4g_*(N)} \right),$$

and defining it arbitrarily off that event, one has

$$\hat{\sigma}_i^{\text{BBP}}(N) - \sigma_i^*(N) \rightarrow 0 \quad \text{in probability}$$

for each $1 \leq i \leq k$.

Proof. See Appendix 5, Proof of Corollary 3.27.

Remark 3.28. Corollary 3.27 adds persistent-spike information to bulk collapse: a macroscopic $\bar{\sigma}_i > 0$ survives as $g_*(N) \rightarrow 0$, eventually exceeds the BBP threshold, and, under the stated gap condition, keeps its spectral direction; the inverse-BBP map then estimates the spike strength consistently.

Remark 3.29 (Assumption map for the Gaussian section). Among the theorem-like results proved in the Gaussian section, the dependence on the hypotheses is as follows:

1. Theorem 2.2 and Proposition 3.30 are standalone random-matrix statements and do not rely on Assumptions 3.1–3.6.
2. Lemma 3.13 and Corollary 3.14 use only Assumptions 3.1–3.3.
3. The first part of Theorem 3.16 uses only Assumptions 3.1–3.3.

4. Theorem 3.18, Corollary 3.19, and the regularized part of Theorem 3.16 use Assumptions 3.1–3.6; Assumption 3.6 is used only to control $\langle S_2, R_2 \rangle_F$.
5. Theorem 3.20 is a standalone sparse signal-plus-noise statement.
6. Theorem 3.21 uses Assumptions 3.1–3.6 together with the one-sided directional stationarity hypothesis stated in the theorem; its pathwise conclusion additionally uses Definition 3.12.
7. Corollary 3.23 is a consequence of Theorem 3.21.
8. Corollary 3.24 additionally uses Assumption 3.8.
9. Corollary 3.25 is a further consequence of Theorem 3.21 plus the asymptotic gap condition

$$\|R_2^*(N)\|_{\text{op}}/\delta_k(N) \rightarrow 0 \quad \text{in probability.}$$

10. Corollary 3.27 uses Theorem 3.21 together with a persistent-spike assumption $\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0$ and the additional exact finite-rank deformation hypothesis stated in the corollary itself.

Proposition 3.30 (Spectral consequence of an MP-scale admissible random component). *Let $M_N/N \rightarrow c \in (0, \infty)$. Let*

$$W_N = R_N + S_N,$$

where $R_N \in \mathbb{R}^{N \times M_N}$ has i.i.d. $\mathcal{N}(0, g_N/N)$ entries with $g_N \rightarrow g_\infty > 0$, and S_N is independent of R_N with

$$\text{rank}(S_N) = o(\min\{N, M_N\}).$$

Then:

1. *the singular-value ESD ν_{R_N} , defined as in Definition 2.1 over the $\min\{N, M_N\}$ singular values of R_N , converges almost surely to the pushforward of the nonzero covariance-spectrum part of $\mu_{\text{MP}}^{c, g_\infty}$ under $x \mapsto \sqrt{x}$; equivalently, when $c > 1$ one first removes the MP atom at 0 and renormalizes. Its support is*

$$\left[\sqrt{g_\infty} |1 - \sqrt{c}|, \sqrt{g_\infty} (1 + \sqrt{c}) \right];$$

2. *the singular-value ESD of W_N has the same weak limit as ν_{R_N} , because*

$$\sup_{x \in \mathbb{R}} |\nu_{W_N}((-\infty, x]) - \nu_{R_N}((-\infty, x])| \leq \frac{\text{rank}(S_N)}{\min\{N, M_N\}} \rightarrow 0;$$

3. *suppose, in addition, that S_N has fixed rank r , its nonzero singular values $\theta_i(N)$ converge to positive limits, and the pair (R_N, S_N) satisfies the finite-rank deformation hypotheses of Theorem 5.4 with deterministic edge $b_+(N)$, critical threshold $\theta_c(N)$, and inverse- D outlier map. Then every spike for which $\mathbb{P}(\theta_i(N) > \theta_c(N) + \gamma) \rightarrow 1$ for some $\gamma > 0$ produces an outlier singular value $s_i(W_N)$ satisfying*

$$s_i(W_N) - D_N^{-1} \left(\frac{1}{\theta_i(N)^2} \right) \rightarrow 0 \quad \text{in probability,}$$

and the inverse- D estimator from Theorem 5.4 recovers $\theta_i(N)$ in probability. In the square Gaussian case $c = 1$, this reduces to the usual BBP outlier transition.

Thus any admissible random component that remains at MP scale leaves a persistent MP-type bulk in the singular spectrum.

Proof. See Appendix 5, Proof of Proposition 3.30.

In this stylized additive model, Theorem 3.18 and Proposition 3.30 say that an MP-scale R_2 leaves a persistent singular-value bulk, and removing it collapses that bulk while lowering the regularized objective by $\lambda_{\text{reg}}\|R_2\|_F^2$, up to the perturbation terms. Theorem 3.21 further rules out admissible sub-MP Frobenius noise at locally stationary limit points, given local-path convergence.

4 Proofs of the perturbation lemmas

Lemma 4.1 (Slack removal). *Let X_N and Y_N be real random variables. Suppose that for every fixed $\eta > 0$,*

$$(X_N - Y_N - \eta)_+ \rightarrow 0 \quad \text{in probability.}$$

Then

$$(X_N - Y_N)_+ \rightarrow 0 \quad \text{in probability.}$$

Equivalently, if $X_N \leq Y_N + \eta + o_{\mathbb{P}}(1)$ for every fixed $\eta > 0$, then $X_N \leq Y_N + o_{\mathbb{P}}(1)$.

Proof. Fix $\delta > 0$ and set $\eta = \delta/2$. Then

$$\mathbb{P}(X_N - Y_N > \delta) \leq \mathbb{P}(X_N - Y_N - \eta > \delta - \eta) \leq \mathbb{P}((X_N - Y_N - \eta)_+ > \delta/2),$$

and the last probability tends to 0 by assumption. $\square$

4.1 Proofs of the deterministic pruning certificates

Proof of Lemma 5.1. For each $s \in T$, the definition of L_s gives

$$\|z_{A_\theta}(s) - z_{A_1}(s)\|_\infty \leq \|\psi_2(A_\theta\psi_1(s)) - \psi_2(A_1\psi_1(s))\|_\infty \leq (1 - \theta)L_s\|R\psi_1(s)\|_\infty.$$

For the single-sample cross-entropy

$$\ell_s(z) := -z_{C(s)} + \log\left(\sum_{j=1}^K e^{z_j}\right),$$

we have $\nabla\ell_s(z) = \rho(z) - e_{C(s)}$, hence $\|\nabla\ell_s(z)\|_1 \leq 2$. Therefore ℓ_s is 2-Lipschitz with respect to $\|\cdot\|_\infty$. Averaging over $s \in T$ yields

$$|\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1)| \leq \frac{2}{|T|} \sum_{s \in T} (1 - \theta)L_s\|R\psi_1(s)\|_\infty = (1 - \theta)B_T(R).$$

$\square$

Proof of Corollary 5.2. The logit perturbation bound in the proof of Lemma 5.1 gives

$$\|z_{A_\theta}(s) - z_{A_1}(s)\|_\infty \leq \eta_s(\theta).$$

If the predicted label changes, then the original top logit and some competing logit must be separated by at most $2\eta_s(\theta)$; otherwise every competing logit remains below the original top logit after perturbation. This proves the prediction-margin claim.

For accuracy, correctness is determined by the sign of the true-label margin. The true class logit can move by at most $\eta_s(\theta)$, and the maximum competing logit can move by at most $\eta_s(\theta)$, so the true-label margin can change by at most $2\eta_s(\theta)$. Therefore the sign of $\Delta_{A_1}^{\text{true}}(s)$ can change only if $|\Delta_{A_1}^{\text{true}}(s)| \leq 2\eta_s(\theta)$. Counting such samples gives the displayed bound. $\square$

Proof of Theorem 5.3. By Lemma 5.1,

$$\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1) \leq (1 - \theta)B_T(R).$$

For the Frobenius term,

$$\|S + \theta R\|_F^2 - \|S + R\|_F^2 = -(1 - \theta^2)\|R\|_F^2 - 2(1 - \theta)\langle S, R \rangle_F.$$

Multiplying this identity by λ_2 and adding the cross-entropy bound gives the displayed path inequality. The η -version follows from

$$-2\lambda_2(1 - \theta)\langle S, R \rangle_F \leq 2\lambda_2(1 - \theta)|\langle S, R \rangle_F|.$$

The full-removal sufficient condition is the case $\theta = 0$. □

Proof of Theorem 5.4. By Lemma 5.1,

$$\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1) \leq (1 - \theta)B_T(R).$$

Since $S \odot R = 0$,

$$\|S + \theta R\|_F^2 = \|S\|_F^2 + \theta^2\|R\|_F^2, \quad \|S + \theta R\|_1 = \|S\|_1 + \theta\|R\|_1.$$

Similarly,

$$\|A_1\|_F^2 = \|S\|_F^2 + \|R\|_F^2, \quad \|A_1\|_1 = \|S\|_1 + \|R\|_1.$$

Substituting these identities into $\mathcal{J}(A_\theta) - \mathcal{J}(A_1)$ gives

$$\mathcal{J}(A_\theta) - \mathcal{J}(A_1) \leq (1 - \theta)B_T(R) - \lambda_2(1 - \theta^2)\|R\|_F^2 - \lambda_1(1 - \theta)\|R\|_1.$$

Taking $\theta = 0$ proves the full-pruning statement and the sufficient decrease condition follows immediately. □

Proof of Theorem 5.5. Apply Theorem 5.4 to the mask decomposition

$$A = (S + Q) + P.$$

The pairwise disjoint-support assumption implies $(S + Q) \odot P = 0$, so the theorem applies with structured part $S + Q$ and removed part P . This gives the displayed path inequality and the full prune-restore statement by taking $\theta = 0$. The sufficient decrease condition follows immediately. □

Proof of Corollary 5.6. The all-pruned upper bound is Theorem 5.4 applied to

$$A = S + R, \quad R = Q + P.$$

The restored upper bound is Theorem 5.5. Since $Q \odot P = 0$,

$$\|R\|_F^2 = \|Q\|_F^2 + \|P\|_F^2, \quad \|R\|_1 = \|Q\|_1 + \|P\|_1.$$

Therefore the restored upper bound is smaller than the all-pruned upper bound exactly when

$$B_T(P) - \lambda_2\|P\|_F^2 - \lambda_1\|P\|_1 < B_T(R) - \lambda_2\|R\|_F^2 - \lambda_1\|R\|_1,$$

which is equivalent to

$$B_T(R) - B_T(P) > \lambda_2\|Q\|_F^2 + \lambda_1\|Q\|_1.$$

□

Proof of Corollary 1.1. The proof is the weighted coordinatewise version of Theorem 5.4. The same cross-entropy bound applies, and $S \odot R = 0$ gives

$$(S_{ij} + \theta R_{ij})^2 = (S_{ij})^2 + \theta^2 (R_{ij})^2, \quad |S_{ij} + \theta R_{ij}| = |S_{ij}| + \theta |R_{ij}|.$$

Multiplying by the nonnegative weights and summing yields the displayed inequality; $\theta = 0$ gives the full-pruning condition. $\square$

Proof of Corollary 1.2. Set $\theta = 1 - h$ in Corollary 1.1 and divide by $h > 0$. This gives

$$\frac{\mathcal{J}_\Gamma(A_{1-h}) - \mathcal{J}_\Gamma(A_1)}{h} \leq B_T(R) - (2-h) \sum_{i,j} \gamma_{2,ij} R_{ij}^2 - \sum_{i,j} \gamma_{1,ij} |R_{ij}|.$$

Taking $h \downarrow 0$ and using one-sided stationarity yields

$$-\varepsilon \leq B_T(R) - 2 \sum_{i,j} \gamma_{2,ij} R_{ij}^2 - \sum_{i,j} \gamma_{1,ij} |R_{ij}|,$$

which is the directional claim. If the pathwise lower bound holds for every θ , combine it with Corollary 1.1 and divide by $1 - \theta$ to obtain the finite- θ inequality. $\square$

Proof of Theorem 5.7. This is Corollary 1.2 with $\gamma_{2,ij} \equiv \lambda_2$ and $\gamma_{1,ij} \equiv \lambda_1$. $\square$

Proof of Theorem 5.8. Apply Theorem 5.4 with $\theta = 0$ at each pruning step in the network obtained after the previous $j - 1$ pruning operations. If $\alpha^{(j)}$ denotes the parameters after step j , then

$$\mathcal{J}(\alpha^{(j)}) - \mathcal{J}(\alpha^{(j-1)}) \leq -\lambda_{2,j} \|R_j\|_F^2 - \lambda_{1,j} \|R_j\|_1 + B_j(R_j).$$

Summing over $j = 1, \dots, M$ telescopes the left-hand side and proves the displayed multi-layer bound. $\square$

Proof of Corollary 1.3. For each $s \in T$, telescoping the logits gives

$$\|z^{(M)}(s) - z^{(0)}(s)\|_\infty \leq \sum_{j=1}^M \|z^{(j)}(s) - z^{(j-1)}(s)\|_\infty \leq \eta_s^{\text{tot}}.$$

The cross-entropy bound follows from the 2-Lipschitz estimate above, and the prediction and accuracy bounds follow by the margin argument of Corollary 5.2 with $\eta_s(\theta)$ replaced by η_s^{tot} . $\square$

Proof of Theorem 5.9. Fix $h \in (0, 1]$, and let $\alpha_h^{(j)}$ be the network after the first j fractional pruning steps, with $\alpha_h^{(0)} = \alpha^{(0)}$ and $\alpha_h^{(M)} = \alpha_h$. At step j , apply the one-layer deterministic certificate with $\theta = 1 - h$ in the intermediate network $\alpha_h^{(j-1)}$. By the assumed validity of the fractional budget $B_j(R_j)$,

$$\mathcal{J}(\alpha_h^{(j)}) - \mathcal{J}(\alpha_h^{(j-1)}) \leq h B_j(R_j) - \lambda_{2,j} (2h - h^2) \|R_j\|_F^2 - \lambda_{1,j} h \|R_j\|_1.$$

Summing over $j = 1, \dots, M$ gives

$$\mathcal{J}(\alpha_h) - \mathcal{J}(\alpha^{(0)}) \leq h \sum_{j=1}^M B_j(R_j) - h(2-h) \sum_{j=1}^M \lambda_{2,j} \|R_j\|_F^2 - h \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1.$$

After division by h , the one-sided directional stationarity assumption and the limit $h \downarrow 0$ imply

$$-\varepsilon \leq \sum_{j=1}^M B_j(R_j) - 2 \sum_{j=1}^M \lambda_{2,j} \|R_j\|_F^2 - \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1,$$

which is the first claim. If the stronger pathwise lower bound holds for every $h \in [0, 1]$, combine it with the displayed upper bound and divide by h to obtain the finite- h inequality. $\square$

Proof of Corollary 5.10. Condition on the vectors $v_s = \psi_1(s)$ and on the bounds L_s . For each $s \in T$ and row i ,

$$\langle r_i, v_s \rangle \sim N\left(0, \frac{g}{n} \|v_s\|_2^2\right).$$

Thus

$$\mathbb{P}\left(|\langle r_i, v_s \rangle| > \|v_s\|_2 \sqrt{\frac{2g}{n} \log \frac{2n|T|}{\delta}}\right) \leq \frac{\delta}{n|T|}.$$

A union bound over $i = 1, \dots, n$ and $s \in T$ gives, with probability at least $1 - \delta$,

$$\|Rv_s\|_\infty \leq \|v_s\|_2 \sqrt{\frac{2g}{n} \log \frac{2n|T|}{\delta}} \quad \forall s \in T.$$

Substitution into (8) proves the claim. $\square$

Proof of Corollary 5.11. For an $n \times m$ Gaussian matrix with entries $N(0, g/n)$, the nominal MP singular-value edge is

$$\sigma_{+,R} = \sqrt{g} \left(1 + \sqrt{\frac{m}{n}}\right).$$

Equivalently,

$$\sqrt{g} = \frac{\sigma_{+,R}}{1 + \sqrt{m/n}}.$$

Substituting this identity into the bound of Corollary 5.10 gives the displayed inequality. $\square$

Proof of Corollary 1.5. Apply Corollary 5.10 at pruning step j with failure probability δ_j , conditional on the intermediate vectors $v_{j,s}$ and Lipschitz bounds $L_{j,s}$. This gives the displayed bound for $B_j(R_j)$ with conditional probability at least $1 - \delta_j$. A union bound over $j = 1, \dots, M$ gives simultaneous validity with probability at least $1 - \sum_j \delta_j \geq 1 - \delta$. Substituting these simultaneous budget bounds into Theorem 5.8 proves the objective inequality. $\square$

Proof of Corollary 1.6. The proof is identical to Corollary 1.5, using Corollary 1.4 in place of Corollary 5.10. $\square$

Proof of Corollary 1.4. Condition on the variables with respect to which v_s and L_s are measurable. Apply the assumed tail bound with

$$t_s := \|v_s\|_2 \sqrt{\frac{g}{cn} \log \frac{2n|T|}{\delta}}.$$

The assumed tail bound and the same row/sample union bound as above give

$$\|Rv_s\|_\infty \leq t_s \quad \forall s \in T$$

with probability at least $1 - \delta$. Substitution into (8) proves the corollary. $\square$

4.2 Proof of Lemma 3.13

Proof. Let

$$\nu := \lambda(W_1 s).$$

Since λ is either absolute value or ReLU, it is coordinatewise 1-Lipschitz and satisfies $\|\nu\|_2 \leq \|W_1 s\|_2$. Write

$$X(s, \alpha_{W_2}) = \lambda(W_3 \lambda((S_2 + R_2)\nu)), \quad X(s, \alpha_{S_2}) = \lambda(W_3 \lambda(S_2 \nu)).$$

Using the 1-Lipschitz property of λ and $\|Av\|_\infty \leq \|A\|_\infty \|v\|_\infty$, we obtain

$$\|X(s, \alpha_{W_2}) - X(s, \alpha_{S_2})\|_\infty \leq \|W_3\|_\infty \|R_2 \nu\|_\infty.$$

So it suffices to bound $\|R_2 \nu\|_\infty$.

By Assumption 3.3, R_2 is independent of W_1 , hence independent of ν . Conditioned on ν , each coordinate $(R_2 \nu)_j$ is centered Gaussian with variance

$$\text{Var}((R_2 \nu)_j \mid \nu) = \frac{g(t)}{N} \|\nu\|_2^2 \leq \frac{\|W_1 s\|_2^2}{N}.$$

Set

$$u_N := 2\sqrt{2} \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}}.$$

If $\|W_1 s\|_2 = 0$, then $\nu = 0$ and the claim is immediate. Otherwise, the elementary Gaussian tail bound gives, conditionally on ν ,

$$\mathbb{P}(|(R_2 \nu)_j| > u_N \mid \nu) \leq 2 \exp\left(-\frac{u_N^2}{2\|W_1 s\|_2^2/N}\right) = 2(2N)^{-4}.$$

A union bound over $j = 1, \dots, N$ gives

$$\mathbb{P}\left(\|R_2 \nu\|_\infty > 2\sqrt{2} \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}} \mid \nu\right) \leq \frac{1}{8N^3} \leq \frac{1}{N}.$$

The same bound holds unconditionally after averaging over ν . Multiplying by $\|W_3\|_\infty$ yields the desired estimate. $\square$

4.3 Proof of Lemma 4.4

Proof. Let $u := \psi_1(s)$. Since

$$X(s, \alpha_W) = \psi_2((R + S)u), \quad X(s, \alpha_S) = \psi_2(Su),$$

Assumption 4.1 gives

$$\|X(s, \alpha_W) - X(s, \alpha_S)\|_\infty \leq L_{\psi_2}(s) \|Ru\|_\infty.$$

Applying Assumption 4.2 with $v = u$ and then averaging over the conditioning variables gives

$$\mathbb{P}(\|Ru\|_\infty > d_1(N)J(u)) \leq d_2(N).$$

Therefore

$$\mathbb{P}(\|X(s, \alpha_W) - X(s, \alpha_S)\|_\infty \leq d_1(N)J(\psi_1(s))L_{\psi_2}(s)) \geq 1 - d_2(N),$$

which is the claim. $\square$

4.4 Justification of Example 2.6

The standard sub-Gaussian tail bound gives absolute constants $c, C > 0$ such that, conditionally on S, ψ_1, ψ_2 ,

$$\mathbb{P}(|\langle r_j, v \rangle| > u \mid S, \psi_1, \psi_2) \leq 2 \exp\left(-c \frac{Nu^2}{\kappa^2 \|v\|_2^2}\right).$$

Taking

$$u = C\kappa\|v\|_2 \sqrt{\frac{\log(2N/\delta)}{N}}$$

with C large enough and applying a union bound over $j = 1, \dots, N$ yields the displayed ℓ_∞ concentration bound.

4.5 Proof of Lemma 3.15

Proof. For $s \in T$, let $X(\alpha, s) \in \mathbb{R}^K$ denote the vector of pre-softmax logits. By Lemma 3.13, for each fixed $s \in T$,

$$\mathbb{P}\left(\|X(\alpha_{S_2}, s) - X(\alpha_{W_2}, s)\|_\infty \leq \tau_N(s)\right) \geq 1 - \frac{1}{N},$$

where

$$\tau_N(s) = 2\sqrt{2} \|W_3\|_\infty \|W_1 s\|_2 \sqrt{\frac{\log(2N)}{N}} = 2\sqrt{2} a(N, s).$$

Let $\ell_s(z) = -z_{C(s)} + \log \sum_j e^{z_j}$. As in the proof of Lemma 5.1, ℓ_s is 2-Lipschitz with respect to $\|\cdot\|_\infty$. Applying this with $z = X(\alpha_{S_2}, s)$ and $z' = X(\alpha_{W_2}, s)$, we get

$$|\ell_s(X(\alpha_{S_2}, s)) - \ell_s(X(\alpha_{W_2}, s))| \leq 4\sqrt{2} a(N, s)$$

on the high-probability event from Lemma 3.13. Since T is finite, a union bound over $s \in T$ yields

$$\mathbb{P}\left(|L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})| \leq \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a(N, s)\right) \geq 1 - \frac{|T|}{N},$$

which proves the lemma. $\square$

4.6 Proof of Theorem 3.16

Proof. For each $s \in T_0$, set $\nu_s = \lambda(W_1 s)$. Applying the same conditional Gaussian row bound and union bound as in Lemma 3.13, now over $s \in T_0$ and $j = 1, \dots, N$ with failure probability δ , gives the displayed simultaneous bound on $\|R_2 \nu_s\|_\infty$. Multiplying by $\|W_3\|_\infty$ yields the logit bound, and the cross-entropy bound follows from the 2-Lipschitz estimate for ℓ_s .

For the regularized statement, use the exact identity

$$L(\alpha_{S_2}) - L(\alpha_{W_2}) = L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2}) - \lambda_{\text{reg}} \|R_2\|_F^2 - 2\lambda_{\text{reg}} \langle S_2, R_2 \rangle_F.$$

Conditional on S_2 ,

$$\langle S_2, R_2 \rangle_F \sim \mathcal{N}\left(0, \frac{g(t)}{N} \|S_2\|_F^2\right),$$

and Assumption 3.6 gives

$$\mathbb{P}(|\langle S_2, R_2 \rangle_F| > \eta \mid S_2) \leq 2e^{-N\eta^2/(2B_N^2)}.$$

Intersecting the logit event with $\{|\langle S_2, R_2 \rangle_F| \leq \eta\}$ gives the claimed probability and inequality. $\square$

4.7 Proof of Theorem 3.20

Proof. Condition on S_N when it is random; the remaining probability is over R_N . If $g_N = 0$, then $R_N = 0$ almost surely and the claim is immediate. Assume $g_N > 0$. For every entry,

$$\mathbb{P}(|R_{ij}| > t_{N,\delta}) \leq 2 \exp\left(-\frac{Nt_{N,\delta}^2}{2g_N}\right) = \frac{\delta}{NM_N},$$

and a union bound gives

$$\mathbb{P}\left(\max_{i,j} |R_{ij}| \leq t_{N,\delta}\right) \geq 1 - \delta.$$

On this event, if $(i, j) \notin \text{supp}(S_N)$, then $|W_{ij}| = |R_{ij}| \leq t_{N,\delta}$, so $\mathcal{H}_{t_{N,\delta}}$ removes that entry. Thus thresholding creates no false positives. If $|(S_N)_{ij}| > 2t_{N,\delta}$, then

$$|W_{ij}| \geq |(S_N)_{ij}| - |R_{ij}| > 2t_{N,\delta} - t_{N,\delta} = t_{N,\delta},$$

so thresholding keeps that entry. This proves the support sandwich and the missed-support inclusion. On the beta-min event $\mathcal{B}_{N,\delta}$, every element of $\text{supp}(S_N)$ has magnitude larger than $2t_{N,\delta}$, so the support sandwich collapses to exact support recovery. Unconditionally, the exact-recovery failure probability is at most $\delta + \mathbb{P}(\mathcal{B}_{N,\delta}^c) \leq \delta + \gamma$. $\square$

5 Low-rank deformations and proofs of the main theorems

5.1 External random-matrix inputs quoted in the manuscript

The Marchenko–Pastur theorem in Theorem 2.2 and the square Gaussian finite-rank deformation theorem in Theorem 5.1 are used as standard random-matrix inputs. The statements are classical; see, for example, [2] for Marchenko–Pastur theory and [1] for the Benaych-Georges–Nadakuditi outlier theorem. All paper-specific consequences stated after these inputs are proved below.

5.2 Finite-rank deformations of random matrices

Theorem 5.1 (Benaych-Georges–Nadakuditi, square Gaussian case [1]). *Let $W_N = R_N + S_N$, where R_N is an $N \times N$ random matrix with i.i.d. $N(0, 1/N)$ entries and S_N is deterministic, or independent of R_N , with fixed rank r and nonzero singular values $\sigma_1 > \dots > \sigma_r > 0$ independent of N . Then the i -th largest singular value $s_i(W_N)$ converges almost surely to*

$$s_i(W_N) \rightarrow \begin{cases} \sigma_i + \sigma_i^{-1}, & \sigma_i > 1, \\ 2, & \sigma_i \leq 1. \end{cases}$$

In particular, subcritical spikes merge with the bulk edge 2.

Proof. This is a classical cited input; see the source note in Appendix 5.1. The variance- g and varying-scale consequences are proved immediately below.

Corollary 5.2 (Square Gaussian case with variance g/N). *Let $W_N = R_N + S_N$, where R_N is an $N \times N$ random matrix with i.i.d. $N(0, g/N)$ entries for some fixed $g > 0$, and S_N is deterministic, or independent of R_N , with fixed rank r and distinct nonzero singular values $\sigma_1 > \dots > \sigma_r > 0$. Then*

$$s_i(W_N) \rightarrow \begin{cases} \sigma_i + g/\sigma_i, & \sigma_i > \sqrt{g}, \\ 2\sqrt{g}, & \sigma_i \leq \sqrt{g}. \end{cases}$$

Proof. Write

$$W_N = \sqrt{g}(\tilde{R}_N + \tilde{S}_N), \quad \tilde{R}_N := R_N/\sqrt{g}, \quad \tilde{S}_N := S_N/\sqrt{g}.$$

Then $\tilde{R}_N$ has i.i.d. $N(0, 1/N)$ entries, and the nonzero singular values of $\tilde{S}_N$ are $\sigma_i/\sqrt{g}$. Apply Theorem 5.1 and multiply by $\sqrt{g}$. $\square$

Theorem 5.3 (Square Gaussian outlier map with varying spike and noise scales). *Let $W_N = R_N + S_N$, where R_N is an $N \times N$ random matrix with i.i.d. $N(0, g_N/N)$ entries and $g_N \rightarrow g_\infty \in [0, \infty)$. Assume S_N has fixed rank r . Write its nonzero singular values in decreasing order,*

$$\sigma_1(S_N) \geq \dots \geq \sigma_r(S_N) > 0,$$

and assume that for each fixed $1 \leq i \leq r$,

$$\sigma_i(S_N) \rightarrow \bar{\sigma}_i \in (0, \infty) \quad \text{in probability}$$

where $\bar{\sigma}_1 \geq \dots \geq \bar{\sigma}_r > 0$, and that S_N is independent of R_N . Then for each fixed $1 \leq i \leq r$,

$$s_i(W_N) \rightarrow \begin{cases} \bar{\sigma}_i + g_\infty/\bar{\sigma}_i, & \bar{\sigma}_i > \sqrt{g_\infty}, \\ 2\sqrt{g_\infty}, & \bar{\sigma}_i \leq \sqrt{g_\infty}. \end{cases} \quad \text{in probability.}$$

Moreover, when $g_\infty > 0$, the usual BGN alignment conclusions hold for simple separated spikes: supercritical spikes have asymptotically nonzero projection onto the corresponding signal directions, while subcritical spikes have vanishing projection. For repeated or clustered spikes, these statements are understood as subspace-alignment statements. When $g_\infty = 0$, any separated spike cluster has convergent spectral projectors under the Wedin gap condition $\|R_N\|_{\text{op}}/\text{gap}_N \rightarrow 0$.

Proof. Condition on S_N . First suppose $g_\infty > 0$. Then $g_N > 0$ for all large N , and after dividing by $\sqrt{g_N}$ one reduces to the unit-variance square Gaussian finite-rank deformation theorem of Benaych-Georges and Nadakuditi [1]. On events whose probability tends to one, the conditioned spike values are arbitrarily close to their deterministic limits; the BGN outlier map is continuous away from the threshold, and at the threshold both branches give the edge. Thus the conditional deterministic-spike result and Slutsky's theorem give the displayed convergence in probability. The limiting outlier map in the unit-variance scaling is

$$\theta \mapsto \theta + \theta^{-1},$$

so in the original scaling it becomes

$$\sigma \mapsto \sigma + \frac{g_\infty}{\sigma},$$

with critical threshold $\sqrt{g_\infty}$ and edge $2\sqrt{g_\infty}$. The singular-vector and spike-subspace alignment statements in the positive-noise case are exactly the corresponding supercritical/subcritical BGN conclusions.

Now suppose $g_\infty = 0$. Write $R_N = \sqrt{g_N} Z_N$, where Z_N has i.i.d. $N(0, 1/N)$ entries. By the standard Gaussian operator-norm bound, $\|Z_N\|_{\text{op}} = O_{\mathbb{P}}(1)$, hence

$$\|R_N\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

Weyl's singular-value inequality gives, for each fixed i ,

$$|s_i(W_N) - s_i(S_N)| \leq \|R_N\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

Since $s_i(S_N) \rightarrow \bar{\sigma}_i$, we obtain $s_i(W_N) \rightarrow \bar{\sigma}_i$, which is the asserted formula with $g_\infty = 0$. The stated projector convergence in the separated-cluster case follows from Wedin's theorem. $\square$

Theorem 5.4 (Abstract inverse- D spike recovery). *Let $W_N = R_N + S_N$. Assume that for each fixed $1 \leq i \leq r$, the i -th nonzero singular value of S_N is $\theta_i(N) > 0$. Assume further that there exist deterministic quantities $b_+(N)$, $\theta_c(N)$, and a strictly decreasing map*

$$D_N : (b_+(N), \infty) \rightarrow (0, D_N(b_+(N)^+))$$

with the following properties:

1. *on supercritical spike sequences, meaning $\mathbb{P}(\theta_i(N) > \theta_c(N)) \rightarrow 1$, one has*

$$s_i(W_N) - \rho_N(\theta_i(N)) \rightarrow 0 \quad \text{in probability,}$$

where

$$\rho_N(\theta) := D_N^{-1}\left(\frac{1}{\theta^2}\right);$$

2. *on subcritical spike sequences, meaning $\mathbb{P}(\theta_i(N) \leq \theta_c(N)) \rightarrow 1$, one has*

$$s_i(W_N) - b_+(N) \rightarrow 0 \quad \text{in probability;}$$

3. *in the supercritical case, the singular vectors associated with $s_i(W_N)$ have asymptotically nonvanishing projection onto the corresponding singular directions of S_N ;*
4. *for each fixed supercritical spike under consideration, $\mathbb{P}(\theta_i(N) > \theta_c(N)) \rightarrow 1$, $\theta_i(N) \rightarrow \bar{\theta}_i \in (0, \infty)$, and there exist deterministic intervals*

$$I_{i,N} \subset (b_+(N), \infty)$$

and deterministic constants $L_{i,N} < \infty$ such that D_N is $L_{i,N}$ -Lipschitz on $I_{i,N}$ and

$$\begin{aligned} \mathbb{P}(\rho_N(\theta_i(N)) \in I_{i,N}) &\rightarrow 1, & \mathbb{P}(s_i(W_N) \in I_{i,N}) &\rightarrow 1, \\ L_{i,N}|s_i(W_N) - \rho_N(\theta_i(N))| &\rightarrow 0 \quad \text{in probability.} \end{aligned}$$

Define the inverse- D estimator on the event $s_i(W_N) > b_+(N)$, whose probability tends to one for every fixed supercritical spike under consideration, by

$$\hat{\theta}_i^{(D)}(N) := D_N(s_i(W_N))^{-1/2},$$

and define it arbitrarily off this event. Then for every fixed supercritical spike,

$$\hat{\theta}_i^{(D)}(N) - \theta_i(N) \rightarrow 0 \quad \text{in probability.}$$

Proof. On the event $\{s_i(W_N) \in I_{i,N}, \rho_N(\theta_i(N)) \in I_{i,N}\}$, whose probability tends to one, the Lipschitz assumption gives

$$|D_N(s_i(W_N)) - D_N(\rho_N(\theta_i(N)))| \leq L_{i,N}|s_i(W_N) - \rho_N(\theta_i(N))|.$$

Therefore

$$D_N(s_i(W_N)) = D_N(\rho_N(\theta_i(N))) + o_{\mathbb{P}}(1) = \frac{1}{\theta_i(N)^2} + o_{\mathbb{P}}(1).$$

Since $\theta_i(N) \rightarrow \bar{\theta}_i \in (0, \infty)$, the quantity $\theta_i(N)^{-2}$ stays in a compact subinterval of $(0, \infty)$ for all large N . Applying the continuous map $x \mapsto x^{-1/2}$ on that compact interval yields

$$\hat{\theta}_i^{(D)}(N) - \theta_i(N) \rightarrow 0 \quad \text{in probability.}$$

This is the inverse-outlier-map viewpoint underlying D -transform shrinkage methods such as OptShrink [3]. □

5.3 Proof of Proposition 3.30

Proof. Part (i) is the MP law on the covariance scale together with the pushforward $x \mapsto \sqrt{x}$ to the singular-value scale. Because ν_{R_N} averages over only $\min\{N, M_N\}$ singular values, when $c > 1$ one removes the covariance-side zero atom of the full MP law before pushing forward and renormalizes the remaining mass. Part (ii) follows from the rank inequality for empirical spectral distributions: if A, B are matrices of the same dimensions, then

$$\sup_x |\nu_A((-\infty, x]) - \nu_B((-\infty, x])| \leq \frac{\text{rank}(A - B)}{\min\{N, M_N\}}.$$

Applying this with $A = W_N$, $B = R_N$, and $A - B = S_N$ gives the claimed bound. Part (iii) is the rectangular outlier principle encoded by the inverse- D transform in Theorem 5.4; the square Gaussian BBP statement follows as the special case $c = 1$. $\square$

5.4 Proofs of Theorem 3.18, Corollary 3.19, Corollary 3.14, and Theorem 3.21 (Gaussian instances)

Proof of Theorem 3.18. Conditional on $S_2(t)$, Assumption 3.3 gives

$$\langle S_2(t), R_2(t) \rangle_F = \text{tr}(S_2(t)^\top R_2(t)) \sim \mathcal{N}\left(0, \frac{g(t)}{N} \|S_2(t)\|_F^2\right).$$

Since $\|S_2(t)\|_F \leq B_N$ and $g(t) \leq 1$, for every $\eta > 0$,

$$\mathbb{P}(|\langle S_2(t), R_2(t) \rangle_F| > \eta \mid S_2(t)) \leq 2e^{-N\eta^2/(2B_N^2)}. \quad (5)$$

Hence

$$\langle S_2(t), R_2(t) \rangle_F \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty. \quad (6)$$

Write

$$L(\alpha) = L_{\text{reg}}(\alpha) + \lambda_{\text{reg}} \sum_{i=1}^L \|W_i\|_F^2.$$

If $W_2 = R_2 + S_2$ and we replace W_2 by S_2 , then

$$\begin{aligned} L(\alpha_{S_2}) - L(\alpha_{W_2}) &= (L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})) - \lambda_{\text{reg}}(\|W_2\|_F^2 - \|S_2\|_F^2) \\ &= (L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})) - \lambda_{\text{reg}}\|R_2\|_F^2 - 2\lambda_{\text{reg}}\langle S_2, R_2 \rangle_F. \end{aligned} \quad (7)$$

By Lemma 3.15,

$$\mathbb{P}\left(|L_{\text{reg}}(\alpha_{S_2}) - L_{\text{reg}}(\alpha_{W_2})| \leq \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a(N, s)\right) \geq 1 - \frac{|T|}{N}.$$

Intersecting this event with $|\langle S_2, R_2 \rangle_F| \leq \eta$, whose probability is bounded below by $1 - 2e^{-N\eta^2/(2B_N^2)}$, yields the finite- N inequality in the theorem. Letting $N \rightarrow \infty$ and using (6) gives the asymptotic expansion. $\square$

Proof of Corollary 3.19. Let $W_2(\vartheta) = S_2 + \vartheta R_2$. Then

$$\begin{aligned} L(\alpha_{W_2(\vartheta)}) - L(\alpha_{W_2}) &= (L_{\text{reg}}(\alpha_{W_2(\vartheta)}) - L_{\text{reg}}(\alpha_{W_2})) \\ &\quad - \lambda_{\text{reg}}\left((1 - \vartheta^2)\|R_2\|_F^2 + 2(1 - \vartheta)\langle S_2, R_2 \rangle_F\right). \end{aligned} \quad (8)$$

For each $s \in T$, write $\nu_s = \lambda(W_1 s)$. Let E_N^{logit} be the event on which

$$\|W_3\|_\infty \|R_2 \nu_s\|_\infty \leq \tau_N(s) \quad \forall s \in T.$$

By the same samplewise Gaussian estimate as in Lemma 3.13 and a union bound over T ,

$$\mathbb{P}(E_N^{\text{logit}}) \geq 1 - \frac{|T|}{N}.$$

On E_N^{logit} , for every $\vartheta \in [0, 1]$ and every $s \in T$,

$$\|X(s, \alpha_{W_2(\vartheta)}) - X(s, \alpha_{W_2})\|_\infty \leq (1 - \vartheta) \|W_3\|_\infty \|R_2 \nu_s\|_\infty \leq (1 - \vartheta) \tau_N(s).$$

The 2-Lipschitz property of the cross-entropy therefore gives, on the same event,

$$|L_{\text{reg}}(\alpha_{W_2(\vartheta)}) - L_{\text{reg}}(\alpha_{W_2})| \leq \frac{4\sqrt{2}(1 - \vartheta)}{|T|} \sum_{s \in T} a(N, s)$$

for every $\vartheta \in [0, 1]$. Combining E_N^{logit} with $|\langle S_2, R_2 \rangle_F| \leq \eta$ proves the finite- N bound. The asymptotic expansion follows from (8) and (6). Since the finite- N event is uniform in ϑ , the same expansion also holds for deterministic sequences $\vartheta_N \in [0, 1]$. $\square$

Proof of Corollary 3.14. By Lemma 3.13, for each fixed $s \in T'$,

$$\mathbb{P}(\|X(s, \alpha_{W_2}) - X(s, \alpha_{S_2})\|_\infty \leq \tau_N(s)) \geq 1 - \frac{1}{N}.$$

A union bound over $s \in T'$ gives simultaneous vector control with probability at least $1 - \frac{|T'|}{N}$. On that event,

$$|\delta X(s, \alpha_{W_2}) - \delta X(s, \alpha_{S_2})| \leq 2\tau_N(s) \quad \forall s \in T'.$$

Therefore labels can only change on samples with $|\delta X(s, \alpha_{W_2})| \leq 2\tau_N(s)$, which yields the claim. $\square$

Proof of Theorem 3.21. Set

$$b_N := \frac{4\sqrt{2}}{|T|} \sum_{s \in T} a_*(N, s) \rightarrow 0.$$

Choose a deterministic sequence $\eta_N \downarrow 0$ such that

$$\frac{N\eta_N^2}{B_N^2} \rightarrow \infty;$$

such a sequence exists because $B_N = o(\sqrt{N})$. By Corollary 3.19 applied at the limit point, there exists an event E_N with $\mathbb{P}(E_N) \rightarrow 1$ such that on E_N , for every $\vartheta \in [0, 1]$,

$$L(\alpha_{\vartheta}^*(N)) - L(\alpha^*(N)) \leq -\lambda_{\text{reg}}(1 - \vartheta^2) \|R_2^*(N)\|_F^2 + (1 - \vartheta)b_N + 2\lambda_{\text{reg}}(1 - \vartheta)\eta_N$$

Write $\vartheta = 1 - h$. Dividing by $h > 0$ gives

$$\frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \leq -\lambda_{\text{reg}}(2 - h) \|R_2^*(N)\|_F^2 + b_N + 2\lambda_{\text{reg}}\eta_N$$

for every $h \in (0, 1]$. On the high-probability event where the one-sided directional stationarity hypothesis also holds, taking $\liminf_{h \downarrow 0}$ yields

$$2\lambda_{\text{reg}} \|R_2^*(N)\|_F^2 \leq b_N + \epsilon_N + 2\lambda_{\text{reg}}\eta_N.$$

Therefore

$$\|R_2^*(N)\|_F^2 \leq \frac{b_N + \epsilon_N}{2\lambda_{\text{reg}}} + \eta_N + o_{\mathbb{P}}(1).$$

Since $\eta_N \rightarrow 0$, this proves the quantitative estimate in the theorem. The vanishing-in-probability statement follows immediately when $b_N + \epsilon_N \rightarrow 0$ in probability.

Now assume the local-path convergence hypothesis from Definition 3.12. Fix $\epsilon > 0$. For each fixed N ,

$$\|R_2(t, N)\|_F \rightarrow \|R_2^*(N)\|_F \quad \text{in probability as } t \rightarrow \infty$$

by the continuous mapping theorem. Since $R_2^*(N)$ is Gaussian (possibly degenerate if $g_*(N) = 0$), the random variable $\|R_2^*(N)\|_F$ has no atom at any positive ϵ . Hence

$$\mathbb{P}(\|R_2(t, N)\|_F \leq \epsilon) \rightarrow \mathbb{P}(\|R_2^*(N)\|_F \leq \epsilon) \quad \text{as } t \rightarrow \infty.$$

Letting $N \rightarrow \infty$ and using the limit-point statement proves the theorem. $\square$

Proof of Corollary 3.23. Suppose, toward a contradiction, that there exist $\eta_0 > 0$ and a subsequence such that $g_*(N)N \geq \eta_0$. Since

$$\|R_2^*(N)\|_F^2 = \frac{g_*(N)}{N} \sum_{i,j=1}^N Z_{ij}^2, \quad Z_{ij} \stackrel{\text{i.i.d.}}{\sim} N(0, 1),$$

we have

$$\frac{\|R_2^*(N)\|_F^2}{g_*(N)N} = \frac{1}{N^2} \sum_{i,j=1}^N Z_{ij}^2 \rightarrow 1 \quad \text{in probability.}$$

Therefore

$$\mathbb{P}(\|R_2^*(N)\|_F^2 \geq \eta_0/2) \rightarrow 1,$$

contradicting Theorem 3.21. Hence $g_*(N)N \rightarrow 0$. The edge statement follows from the square-case MP singular-value edge $2\sqrt{g_*(N)}$. In the rectangular $N \times M_N$ case,

$$\|R_2^*(N)\|_F^2 = \frac{g_*(N)}{N} \sum_{i=1}^N \sum_{j=1}^{M_N} Z_{ij}^2, \quad \frac{1}{NM_N} \sum_{i,j} Z_{ij}^2 \rightarrow 1$$

in probability. Thus $\|R_2^*(N)\|_F \rightarrow 0$ forces $g_*(N)M_N \rightarrow 0$, and if $M_N/N \rightarrow c \in (0, \infty)$ this is equivalent to $g_*(N)N \rightarrow 0$. $\square$

Proof of Corollary 3.24. Since $\|R_2^*(N)\|_{\text{op}} \leq \|R_2^*(N)\|_F$, Theorem 3.21 immediately gives

$$\|R_2^*(N)\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

This proves (i).

For (ii), singular-value Weyl perturbation gives

$$|s_i(W_2^*(N)) - s_i(S_2^*(N))| \leq \|R_2^*(N)\|_{\text{op}} \quad \forall i.$$

Hence for each fixed $i \geq 1$,

$$|s_i(W_2^*(N)) - \sigma_i^*(N)| \leq \|R_2^*(N)\|_{\text{op}} \rightarrow 0$$

in probability.

Let $S_{2,\leq k_N}^*(N)$ denote the best rank- k_N approximation of $S_2^*(N)$. By the Eckart–Young theorem,

$$\begin{aligned} \sum_{i>k_N} s_i(W_2^*(N))^2 &= \inf_{\text{rank}(A)\leq k_N} \|W_2^*(N) - A\|_F^2 \leq \|W_2^*(N) - S_{2,\leq k_N}^*(N)\|_F^2 \\ &= \|R_2^*(N) + (S_2^*(N) - S_{2,\leq k_N}^*(N))\|_F^2 \leq 2\|R_2^*(N)\|_F^2 + 2 \sum_{i>k_N} \sigma_i^*(N)^2, \end{aligned}$$

which proves (iii) using Theorem 3.21 and Assumption 3.8.

For (iv), fix $\epsilon > 0$. Since $\|W_2^*(N)\|_F \leq \|S_2^*(N)\|_F + \|R_2^*(N)\|_F$, we have

$$\nu_{W_2^*(N)}((\epsilon, \infty)) \leq \frac{\|W_2^*(N)\|_F^2}{N\epsilon^2} \leq \frac{2B_N^2 + 2\|R_2^*(N)\|_F^2}{N\epsilon^2},$$

which converges to 0 in probability because $B_N = o(\sqrt{N})$ and $\|R_2^*(N)\|_F \rightarrow 0$ in probability. The final claim follows from (ii). $\square$

Proof of Corollary 3.25. By Corollary 3.24(i),

$$\|R_2^*(N)\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

Hence the event

$$F_N := \left\{ \|R_2^*(N)\|_{\text{op}} \leq \frac{\delta_k(N)}{4} \right\}$$

satisfies $\mathbb{P}(F_N) \rightarrow 1$.

On F_N , Weyl's singular-value inequality gives

$$s_k(W_2^*(N)) \geq \sigma_k^*(N) - \|R_2^*(N)\|_{\text{op}},$$

while

$$s_{k+1}(W_2^*(N)) \leq \sigma_{k+1}^*(N) + \|R_2^*(N)\|_{\text{op}},$$

with the convention $\sigma_j^*(N) = 0$ beyond the rank of $S_2^*(N)$. Therefore,

$$s_k(W_2^*(N)) - s_{k+1}(W_2^*(N)) \geq \delta_k(N) - 2\|R_2^*(N)\|_{\text{op}} \geq \frac{\delta_k(N)}{2}$$

on F_N . Thus on F_N , the top- k singular subspace of $W_2^*(N)$ is separated from the rest by a positive gap.

Applying Wedin's $\sin \Theta$ theorem for singular subspaces [5, 4] to

$$W_2^*(N) = S_2^*(N) + R_2^*(N),$$

with target rank k , yields a universal constant $C > 0$ such that on F_N ,

$$\|\hat{P}_{U,k}(N) - P_{U^*,k}(N)\|_{\text{op}} \leq C \frac{\|R_2^*(N)\|_{\text{op}}}{\delta_k(N)}, \quad \|\hat{P}_{V,k}(N) - P_{V^*,k}(N)\|_{\text{op}} \leq C \frac{\|R_2^*(N)\|_{\text{op}}}{\delta_k(N)}.$$

Since $\|R_2^*(N)\|_{\text{op}}/\delta_k(N) \rightarrow 0$ in probability and $\mathbb{P}(F_N) \rightarrow 1$, both projector differences converge to 0 in probability. $\square$

Proof of Corollary 3.27. By Corollary 3.23,

$$g_*(N)N \rightarrow 0,$$

hence $g_*(N) \rightarrow 0$. Since $\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0$ in probability for each $1 \leq i \leq k$,

$$\mathbb{P}(\sigma_i^*(N) > \sqrt{g_*(N)}) \rightarrow 1,$$

which proves (i).

By Corollary 3.24(i),

$$\|R_2^*(N)\|_{\text{op}} \rightarrow 0 \quad \text{in probability.}$$

Hence Weyl's inequality gives

$$s_i(W_2^*(N)) - \sigma_i^*(N) \rightarrow 0 \quad \text{in probability}$$

for each fixed $1 \leq i \leq k$. Since $\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0$ and $2\sqrt{g_*(N)} \rightarrow 0$, these singular values lie above the Gaussian bulk edge $2\sqrt{g_*(N)}$ with probability tending to one. Equivalently, in the exact finite-rank Gaussian deformation language of Theorem 5.3, the persistent spikes are eventually supercritical. If the additional gap condition in (ii) holds for a spike, Wedin's theorem applied to $W_2^*(N) = S_2^*(N) + R_2^*(N)$ gives, for the associated left and right rank-one projectors,

$$\|\hat{P}_i(N) - P_i^*(N)\|_{\text{op}} \leq C \frac{\|R_2^*(N)\|_{\text{op}}}{\Delta_i(N)} \rightarrow 0 \quad \text{in probability,}$$

and similarly for the right projector. This proves the strengthened projector statement in (ii).

Set $y_N = s_i(W_2^*(N))$. The preceding Weyl estimate gives $y_N - \sigma_i^*(N) \rightarrow 0$ in probability, hence $y_N \rightarrow \bar{\sigma}_i > 0$. Also $g_*(N) \rightarrow 0$, so the event $y_N^2 > 4g_*(N)$ has probability tending to one. On this event,

$$0 \leq y_N - \hat{\sigma}_i^{\text{BBP}}(N) = \frac{1}{2} \left(y_N - \sqrt{y_N^2 - 4g_*(N)} \right) = \frac{2g_*(N)}{y_N + \sqrt{y_N^2 - 4g_*(N)}} \rightarrow 0$$

in probability. Therefore

$$\hat{\sigma}_i^{\text{BBP}}(N) - \sigma_i^*(N) \rightarrow 0 \quad \text{in probability,}$$

which proves (iii). □

5.5 Proofs of the abstract-generalization corollaries and theorems

The results below are the abstract additive-perturbation analogues of the Gaussian-instance proofs of Appendix 5. Each is obtained by substituting the abstract logit-perturbation lemma (Lemma 4.4) for the Gaussian logit perturbation bound (Lemma 3.13) and applying the abstract Assumptions 4.1–4.3 in place of Gaussian Assumptions 3.3–3.6.

Proof of Theorem 4.5. For $s \in T$, let $X(\alpha, s)$ denote the vector of pre-softmax logits. By Lemma 4.4, for each fixed $s \in T$,

$$\mathbb{P}(\|X(\alpha_S, s) - X(\alpha_W, s)\|_{\infty} \leq a_{\phi}(N, s)) \geq 1 - d_2(N),$$

where $a_{\phi}(N, s) = h_{\phi}(s)d_1(N)$. The single-sample cross-entropy is 2-Lipschitz in $\|\cdot\|_{\infty}$, so on this event $|\ell_s(X(\alpha_S, s)) - \ell_s(X(\alpha_W, s))| \leq 2a_{\phi}(N, s)$. A union bound over $s \in T$ gives

$$\mathbb{P}\left(|L_{\text{reg}}(\alpha_S) - L_{\text{reg}}(\alpha_W)| \leq \frac{2}{|T|} \sum_{s \in T} a_{\phi}(N, s)\right) \geq 1 - |T|d_2(N).$$

Since $a_\phi(N, s) \rightarrow 0$, $L_{\text{reg}}(\alpha_S) - L_{\text{reg}}(\alpha_W) \rightarrow 0$ in probability. Combined with the exact regularizer identity $\|W\|_F^2 - \|S\|_F^2 = \|R\|_F^2 + 2\langle S, R \rangle_F = \|R\|_F^2 + o_{\mathbb{P}}(1)$, this gives $L(\alpha_S) = L(\alpha_W) - \lambda_{\text{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1)$. $\square$

Proof of Corollary 2.3. Let E_N^{logit} be the samplewise logit event $\|X(\alpha_S, s) - X(\alpha_W, s)\|_\infty \leq a_\phi(N, s)$ for all $s \in T$. By Lemma 4.4 and a union bound, $\mathbb{P}(E_N^{\text{logit}}) \geq 1 - |T|d_2(N)$. On this event the samplewise perturbation argument with $(1 - \vartheta)R$ in place of R gives, for every $\vartheta \in [0, 1]$,

$$|L_{\text{reg}}(\alpha_{W(\vartheta)}) - L_{\text{reg}}(\alpha_W)| \leq (1 - \vartheta) \frac{2}{|T|} \sum_{s \in T} a_\phi(N, s).$$

Let $E_N^{\text{cross}}(\eta) := \{|\langle S, R \rangle_F| \leq \eta\}$ and $E_{N, \eta} := E_N^{\text{logit}} \cap E_N^{\text{cross}}(\eta)$. Then $\mathbb{P}(E_{N, \eta}) \geq 1 - |T|d_2(N) - \mathbb{P}(|\langle S, R \rangle_F| > \eta)$. On $E_{N, \eta}$, the exact regularizer identity $\|W\|_F^2 - \|W(\vartheta)\|_F^2 = (1 - \vartheta^2)\|R\|_F^2 + 2(1 - \vartheta)\langle S, R \rangle_F$ implies

$$L(\alpha_{W(\vartheta)}) \leq L(\alpha_W) - \lambda_{\text{reg}}(1 - \vartheta^2)\|R\|_F^2 + (1 - \vartheta) \frac{2}{|T|} \sum_{s \in T} a_\phi(N, s) + 2\lambda_{\text{reg}}(1 - \vartheta)\eta$$

for every $\vartheta \in [0, 1]$, which is exactly the claim. $\square$

Proof of Corollary 2.4. Since $\theta_c(N) \rightarrow 0$ and $\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0$ in probability, $\mathbb{P}(\sigma_i^*(N) > \theta_c(N)) \rightarrow 1$ for each $1 \leq i \leq k$, which proves (i). On the same high-probability event, the assumed BGN-type deformation regime places the i -th spike above the critical threshold, so the corresponding singular value cluster of $W^*(N)$ lies above the bulk edge. If the gap condition in (ii) holds for a spike, Wedin's theorem gives $\|\widehat{P}_i(N) - P_i^*(N)\|_{\text{op}} \leq C\|R^*(N)\|_{\text{op}}/\Delta_i(N) \rightarrow 0$ in probability, for the left rank-one projectors (and similarly for the right projectors), proving (ii). Applying Theorem 5.4 with $\theta_i(N) = \sigma_i^*(N)$ then gives $\widehat{\sigma}_i^{(D)}(N) - \sigma_i^*(N) \rightarrow 0$ in probability for each fixed $1 \leq i \leq k$, which is (iii). $\square$

Proof of Corollary 2.1. By Lemma 4.4 and a union bound over $s \in T_N$, with probability at least $1 - |T_N|d_2(N)$,

$$\|X(\alpha_S, s) - X(\alpha_W, s)\|_\infty \leq a_\phi(N, s) \quad \forall s \in T_N.$$

On this event, the 2-Lipschitz property of cross-entropy gives

$$|L_{\text{reg}, T_N}(\alpha_S) - L_{\text{reg}, T_N}(\alpha_W)| \leq \frac{2}{|T_N|} \sum_{s \in T_N} a_\phi(N, s).$$

The right-hand side is $o_{\mathbb{P}}(1)$, and the event has probability tending to one because $|T_N|d_2(N) \rightarrow 0$. The regularized identity

$$\|W\|_F^2 - \|S\|_F^2 = \|R\|_F^2 + 2\langle S, R \rangle_F$$

and the cross-term hypothesis then yield

$$L_{T_N}(\alpha_S) = L_{T_N}(\alpha_W) - \lambda_{\text{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1).$$

$\square$

Proof of Corollary 2.2. Set $W(\vartheta) = S + \vartheta R$. Repeating the argument above with perturbation $(1 - \vartheta)R$ in place of R , we get

$$L_{\text{reg}}(\alpha_{W(\vartheta)}) - L_{\text{reg}}(\alpha_W) = o_{\mathbb{P}}(1).$$

Also,

$$\|W\|_F^2 - \|W(\vartheta)\|_F^2 = (1 - \vartheta^2)\|R\|_F^2 + 2(1 - \vartheta)\langle S, R \rangle_F = (1 - \vartheta^2)\|R\|_F^2 + o_{\mathbb{P}}(1).$$

Hence

$$L(\alpha_{W(\vartheta)}) = L(\alpha_W) - \lambda_{\text{reg}}(1 - \vartheta^2)\|R\|_F^2 + o_{\mathbb{P}}(1).$$

The same argument, or the uniform finite- N pathwise bound of Corollary 2.3, gives the statement along any deterministic sequence $\vartheta_N \in [0, 1]$. $\square$

Proof of Theorem 4.6. Order the pruned layers as $\mathcal{P} = \{\ell_1, \dots, \ell_m\}$. Let $\alpha^{(j)}$ denote the network parameters after the first j replacements, so $\alpha^{(0)}$ is the original network and $\alpha^{(m)} = \alpha^{(|\mathcal{P}|)}$. By assumption, at stage j the hypotheses of Theorem 4.5 apply to layer ℓ_j , hence

$$L(\alpha^{(j)}) = L(\alpha^{(j-1)}) - \lambda_{\text{reg}}\|R_{\ell_j}\|_F^2 + o_{\mathbb{P}}(1).$$

Summing these identities over $j = 1, \dots, m$ and using that $m = |\mathcal{P}|$ is fixed gives

$$L(\alpha^{(m)}) = L(\alpha^{(0)}) - \lambda_{\text{reg}} \sum_{j=1}^m \|R_{\ell_j}\|_F^2 + o_{\mathbb{P}}(1),$$

which is the claimed finite-layer telescoping formula.

For the growing-layer version, use the finite- N single-layer bound on the intersection $\bigcap_{\ell \in \mathcal{P}_N} E_{\ell, N}$, whose probability tends to one by the assumed summability of the failure probabilities. Summing the exact loss identities over the ordered layers gives the accumulated regularization decrease $-\lambda_{\text{reg}} \sum_{\ell \in \mathcal{P}_N} \|R_{\ell}\|_F^2$ plus the sum of the cross-entropy perturbation errors and the accumulated cross term. The two summability hypotheses make these remainders $o_{\mathbb{P}}(1)$, yielding the same formula with $\mathcal{P}_N$. $\square$

Proof of Theorem 4.7. Let $b_N := (2/|T|) \sum_{s \in T} a_{\phi}(N, s) \rightarrow 0$. Since $\langle S^*(N), R^*(N) \rangle_F = o_{\mathbb{P}}(1)$, choose a deterministic sequence $\eta_N \downarrow 0$ such that $\mathbb{P}(|\langle S^*(N), R^*(N) \rangle_F| > \eta_N) \rightarrow 0$. By Corollary 2.3 applied at the limit point, there is an event E_N with $\mathbb{P}(E_N) \rightarrow 1$ such that on E_N , for every $\vartheta \in [0, 1]$,

$$L(\alpha_{\vartheta}^*(N)) - L(\alpha^*(N)) \leq -\lambda_{\text{reg}}(1 - \vartheta^2)\|R^*(N)\|_F^2 + (1 - \vartheta)b_N + 2\lambda_{\text{reg}}(1 - \vartheta)\eta_N.$$

Write $\vartheta = 1 - h$ and divide by $h > 0$:

$$\frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \leq -\lambda_{\text{reg}}(2 - h)\|R^*(N)\|_F^2 + b_N + 2\lambda_{\text{reg}}\eta_N \quad \forall h \in (0, 1].$$

On the high-probability event where the one-sided directional stationarity hypothesis also holds, taking $\liminf_{h \downarrow 0}$ yields $2\lambda_{\text{reg}}\|R^*(N)\|_F^2 \leq b_N + \epsilon_N + 2\lambda_{\text{reg}}\eta_N$, so

$$\|R^*(N)\|_F^2 \leq (b_N + \epsilon_N)/(2\lambda_{\text{reg}}) + \eta_N + o_{\mathbb{P}}(1).$$

Since $\eta_N \rightarrow 0$, this proves the quantitative estimate; the vanishing-in-probability statement follows whenever $b_N + \epsilon_N \rightarrow 0$ in probability. The pathwise conclusion under the convergence/continuity hypotheses follows because $\|R(t, N)\|_F \rightarrow \|R^*(N)\|_F$ in probability and $\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0$. $\square$

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